

Shifted Catalan Hankel Polynomials

Exact coefficients, sharp recurrences,
and a denominator conjecture

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With Part II (28 September 2026): root collisions and sharp recurrences for arbitrary polynomial multipliers

Abstract

For the Catalan numbers C_n , consider the size- N determinant $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})_{0 \leq i, j < N}$. We give a self-contained proof of the denominator-exponent conjecture stated in Paul Barry's 2020 note on this family. With $d = \binom{m}{2}$, the generating function is $P_m(t)/(1 - (a + 2b)t + b^2t^2)^{d+1}$. The denominator is minimal whenever $b \neq 0$, $a \neq 0$, and $a \neq -4b$. We determine every exceptional case, prove that $\deg P_m = (m-1)^2$ for $b \neq 0$, and establish a reciprocal symmetry of the numerator. An explicit positive-integer coefficient formula also identifies and repairs a one-unit shift in the source's separate Conjecture 2. The essential degree reduction is a Vandermonde cancellation in confluent Christoffel minors. It yields an exact sum of two exponentials with polynomial coefficients, a finite full asymptotic expansion, and an explicit first correction. Independent integer and rational computations, including direct determinants and polynomial-identity certificates at fixed sizes, accompany the proofs.

Addition of 28 September 2026 (batch 39). Part II, from a later manuscript, replaces $x^m(a + bx)$ by an arbitrary nonzero multiplier $q(x) = \gamma x^m(x-4)^\ell \prod_i (x - c_i)^{t_i}$. Each exponential sector has a polynomial coefficient of exact degree $\binom{m}{2} + \binom{\ell+1}{2} + \sum_i k_i(t_i - k_i)$ with a nonzero product leading coefficient, so off resonance the minimal recurrence order is $\prod_i (t_i + 1) \left(\binom{m}{2} + \binom{\ell+1}{2} + 1 + \frac{1}{6} \sum_i t_i(t_i - 1) \right)$; a secondary Hankel determinant detects exactly when it drops, and a certified finite-prefix algorithm handles the resonances.

Main result. For all nonexceptional complex parameters,

$$\text{minimal constant-coefficient recurrence order} = m(m-1) + 2.$$

The proof is uniform in the determinant size and the Catalan shift.

Status and scope. This is an unrefereed, AI-assisted mathematical manuscript, not a formalized proof. Its theorems are supported by explicit arguments, not inferred from finite tests. The target is a conjecture in a specified source version; no claim of first discovery or exhaustive verification of present-day open status is made. Classical Christoffel formulas and the known general rationality theory are credited, rather than presented as new.

Reproducibility. The accompanying archive contains this L^AT_EX source, the PDF, a standard-library-only Python implementation, an independently organized verification suite, and exact machine-readable data. No external mathematical software is needed to rerun the tests.

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1 The problem and the contribution

Let

$$C_n = \frac{1}{n+1} \binom{2n}{n} \quad (n \in \mathbb{N}_0) \quad (1.1)$$

be the Catalan numbers. We use determinant size, rather than the last row index, throughout:

$$D_N^{(m)}(a, b) = \det_{0 \leq i, j < N} (aC_{i+j+m} + bC_{i+j+m+1}), \quad D_0^{(m)} = 1. \quad (1.2)$$

Here $m, N \in \mathbb{N}_0$, and the parameters may be indeterminates or complex numbers. Define the ordinary generating function

$$F_m(t; a, b) = \sum_{N \geq 0} D_N^{(m)}(a, b) t^N. \quad (1.3)$$

All generating-function identities below first make sense as identities of formal power series. Assertions about poles concern the associated rational functions.

Barry's preprint [1], version 1, explicitly conjectures on page 4 that the exponent of the quadratic generating-function denominator is $\binom{m}{2} + 1$. The same page contains a more complicated coefficient conjecture. These are distinct statements: the denominator claim is correct in the uniform sense proved here, whereas the printed coefficient formula uses a shift one smaller than its accompanying determinant. Appendix A makes the translation precise.

There is substantial prior theory. Dougherty, French, Saderholm, and Qian [2] studied recurrences for linear combinations of Catalan numbers. Krattenthaler [3] gives a detailed account of the Christoffel-type determinant identity, confluence, and constant-coefficient recurrences; his Section 8 also records Elouafi's prior proof of general Catalan recurrence assertions. For a multiplier of degree r , that general construction supplies an order at most 2^r . Consequently, rationality by itself is *not* the research claim here. For the special multiplier $x^m(a + bx)$, the objective is the exact, much smaller order, together with its sharp parameter classification.

The article separates two useful calculations. Jacobi orthogonality immediately exposes the individual coefficients. A confluent determinant, on the other hand, exposes the dependence on N and explains the precise recurrence order. Their independent implementations provide a useful check against indexing and normalization errors.

1.1 A notation guide

We shall repeatedly use

$$d = \binom{m}{2}, \quad e = d + 1, \quad K = (m - 1)^2, \quad q(t) = 1 - (a + 2b)t + b^2 t^2. \quad (1.4)$$

The shifted pure Catalan determinant and its leading coefficient are

$$H_N^{(m)} = \det_{0 \leq i, j < N} (C_{i+j+m}) = \prod_{1 \leq j \leq i < m} \frac{2N + i + j}{i + j}, \quad (1.5)$$

$$h_m = \prod_{1 \leq j \leq i < m} \frac{2}{i + j}. \quad (1.6)$$

Every empty product and empty determinant equals 1. In particular, $H_N^{(0)} = H_N^{(1)} = 1$ and $h_0 = h_1 = 1$. We prove (1.5) rather than assuming it. As a polynomial in N , $H_N^{(m)}$ has degree d and leading coefficient h_m . The notation $(u)_k = u(u + 1) \cdots (u + k - 1)$ denotes a rising factorial, with $(u)_0 = 1$.

2 Statements of the main theorems

Theorem 2.1 (Explicit coefficients). *For every $N, m \in \mathbb{N}_0$,*

$$D_N^{(m)}(a, b) = \sum_{k=0}^N c_{N,m,k} a^k b^{N-k}, \quad c_{N,m,k} = H_N^{(m+1)} \binom{N}{k} \frac{(N+m+1)_k}{4^k (m+\frac{1}{2})_k}. \quad (2.1)$$

All the coefficients $c_{N,m,k}$ are positive integers. Equivalently,

$$c_{N,m,k} = H_N^{(m+1)} \binom{N}{k} \frac{(N+m+k)!(2m)!(m+k)!}{(N+m)!(2m+2k)!m!}. \quad (2.2)$$

They can be generated without evaluating any determinant:

$$c_{N,m,0} = H_N^{(m+1)}, \quad \frac{c_{N,m,k+1}}{c_{N,m,k}} = \frac{(N-k)(N+m+1+k)}{2(k+1)(2m+2k+1)}. \quad (2.3)$$

Theorem 2.2 (Uniform rational form and sharp denominator). *For every $m \in \mathbb{N}_0$ there is a polynomial $P_m \in \mathbb{Z}[a, b, t]$ such that*

$$F_m(t; a, b) = \frac{P_m(t; a, b)}{q(t)^{d+1}}. \quad (2.4)$$

It satisfies $P_m(0; a, b) = 1$ and $\deg_t P_m \leq K$. If $b \neq 0$, then $\deg_t P_m = K$ and

$$P_m(t; a, b) = s_m (bt)^K P_m\left(\frac{1}{b^2 t}; a, b\right), \quad s_m = (-1)^{(m-1)(m-2)/2}. \quad (2.5)$$

In particular, $[t^K]P_m = s_m b^K$. If $b \neq 0$, $a \neq 0$, and $a \neq -4b$, the representation (2.4) is reduced over $\mathbb{C}[t]$. The sequence has minimal constant-coefficient recurrence characteristic polynomial

$$(X^2 - (a+2b)X + b^2)^{d+1}, \quad (2.6)$$

and hence minimal order $2d+2 = m(m-1)+2$.

Here “minimal” refers even to recurrences allowed to start after a finite initial segment. Thus minimality is not an artifact of the empty-determinant convention. The polynomial numerator in (2.4) remains the *uniform, possibly unreduced* numerator at exceptional parameters.

Theorem 2.3 (Every exceptional denominator). *For fixed complex a, b , the reduced denominator of F_m , normalized to have constant term 1, is exactly the following:*

Parameter condition	Reduced denominator
$a = b = 0$	1
$b = 0, a \neq 0$	$(1 - at)^{\binom{m}{2}+1}$
$a = 0, b \neq 0$	$(1 - bt)^{\binom{m+1}{2}+1}$
$a = -4b, b \neq 0$	$(1 + bt)^{\binom{m}{2}+2}$
$b \neq 0, a(a+4b) \neq 0$	$q(t)^{\binom{m}{2}+1}$

Theorem 2.4 (Finite asymptotic expansion). *Let $a, b > 0$ be fixed and put*

$$r = \frac{a/b + 2 + \sqrt{(a/b)(a/b+4)}}{2} > 1, \quad \lambda_+ = br, \quad \lambda_- = b/r. \quad (2.7)$$

There are explicit degree- d polynomials $B_+(N), B_-(N)$ such that

$$D_N^{(m)}(a, b) = B_+(N)\lambda_+^N + B_-(N)\lambda_-^N \quad (N \geq 0). \quad (2.8)$$

In particular,

$$D_N^{(m)}(a, b) = A_m N^d \lambda_+^N \left(1 + \frac{d \left(\frac{m}{2} - \frac{1}{r-1} \right)}{N} + O(N^{-2}) + O(r^{-2N}) \right), \quad (2.9)$$

$$A_m = h_m \frac{r^{m+1}}{(r-1)^m(r+1)}. \quad (2.10)$$

The entire algebraic expansion terminates after N^{-d} ; the second exponential is also multiplied by a polynomial of degree d . For $d = 0$, there are no algebraic correction terms at all.

2.1 The scale of the improvement

For a degree- $(m+1)$ multiplier the general 2^{m+1} construction need not be minimal. The theorem determines the actual order in this family. The first values are:

m	Exponent e	Exact order $2e$	Numerator degree K	2^{m+1}
0	1	2	1	2
1	1	2	0	4
2	2	4	1	8
3	4	8	4	16
4	7	14	9	32
5	11	22	16	64
6	16	32	25	128
8	29	58	49	512
10	46	92	81	2048

The reduction is not obtained by fitting a shorter recurrence to a long initial segment. It comes from a cancellation of polynomial degree in a family of minors, proved in Section 5.

3 Moments, orthogonality, and the coefficient formula

3.1 Catalan moments and a normalized Jacobi polynomial

For $m \in \mathbb{N}_0$ define a positive measure on $(0, 4)$ by

$$d\mu_m(x) = \frac{1}{2\pi} x^{m-1/2} (4-x)^{1/2} dx. \quad (3.1)$$

The beta integral, after $x = 4t$, gives

$$\int_0^4 x^n d\mu_m(x) = \frac{4^{n+m+1}}{2\pi} B(n+m+\frac{1}{2}, \frac{3}{2}) = C_{n+m}. \quad (3.2)$$

Consequently every moment matrix is positive definite, so its determinant is nonzero and its monic orthogonal polynomials are well-defined.

Lemma 3.1 (The monic polynomial). *The degree- N monic orthogonal polynomial for μ_m is*

$$p_N^{(m)}(x) = (-1)^N \frac{4^N (m + \frac{1}{2})_N}{(N + m + 1)_N} {}_2F_1 \left(\begin{matrix} -N, N + m + 1 \\ m + \frac{1}{2} \end{matrix}; x/4 \right). \quad (3.3)$$

The hypergeometric series terminates after $N + 1$ terms.

Proof. Write $\alpha = m - \frac{1}{2}$, $\beta = \frac{1}{2}$, so both are greater than -1 . A Rodrigues representation on $0 < t < 1$ is

$$\frac{t^{-\alpha}(1-t)^{-\beta}}{(\alpha+1)_N} \frac{d^N}{dt^N} \left(t^{N+\alpha}(1-t)^{N+\beta} \right) = {}_2F_1 \left(\begin{matrix} -N, N + \alpha + \beta + 1 \\ \alpha + 1 \end{matrix}; t \right). \quad (3.4)$$

Leibniz's rule gives the polynomial on the right, including its normalization at $t = 0$. Multiplying the left side by $t^j(1-t)^\beta$ and integrating by parts N times yields zero for $0 \leq j < N$. Every boundary term vanishes: after at most $N - 1$ derivatives, the endpoint exponents that can occur are at least $\alpha + 1 > 0$ and $\beta + 1 > 0$. The final derivative of t^j is zero.

The coefficient of t^N on the right is $(-1)^N (N + \alpha + \beta + 1)_N / (\alpha + 1)_N$. Replacing t by $x/4$ and normalizing this leading coefficient gives (3.3). This also identifies the polynomial with the standard Jacobi family; see the conventional hypergeometric representation in DLMF [4], Eq. 18.5.7. $\square$

3.2 A characteristic-polynomial determinant identity

The following elementary identity will be used twice.

Lemma 3.2. *Let v_j be the moments of a positive measure, let $H_N = \det(v_{i+j})_{0 \leq i, j < N}$, and let p_N be its monic orthogonal polynomial. Then*

$$\det_{0 \leq i, j < N} (v_{i+j+1} - z v_{i+j}) = (-1)^N H_N p_N(z). \quad (3.5)$$

Proof. The monic polynomial is represented by

$$p_N(z) = \frac{1}{H_N} \det \begin{pmatrix} v_0 & v_1 & \cdots & v_N \\ v_1 & v_2 & \cdots & v_{N+1} \\ \vdots & \vdots & & \vdots \\ v_{N-1} & v_N & \cdots & v_{2N-1} \\ 1 & z & \cdots & z^N \end{pmatrix}.$$

Its leading coefficient is one, and integrating it against any power z^j , $j < N$, creates a repeated row. Thus it is indeed p_N . Starting with the last column and moving left, replace col_j by $\text{col}_j - z \text{col}_{j-1}$. The last row becomes $(1, 0, \dots, 0)$. Expansion along it gives (3.5), with sign $(-1)^N$. $\square$

Applying the lemma to μ_m and $z = -a/b$ gives, for $b \neq 0$,

$$D_N^{(m)}(a, b) = (-b)^N H_N^{(m)} p_N^{(m)}(-a/b). \quad (3.6)$$

3.3 A proof of the shifted Catalan product

For $m = 0$, (3.3) reduces to

$$p_n^{(0)}(x) = \sum_{j=0}^n (-1)^{n-j} \binom{n+j}{2j} x^j. \quad (3.7)$$

Its initial values and recurrence are

$$p_0^{(0)} = 1, \quad p_1^{(0)} = x - 1, \quad p_{n+2}^{(0)} = (x - 2)p_{n+1}^{(0)} - p_n^{(0)}. \quad (3.8)$$

These follow directly by comparing the binomial coefficients in (3.7). If η_n denotes the squared norm of $p_n^{(0)}$, the coefficient 1 of $p_{n-1}^{(0)}$ in $x p_n^{(0)}$ implies $\eta_n = \eta_{n-1}$. Indeed, the two ways of writing $\langle x p_n^{(0)}, p_{n-1}^{(0)} \rangle$ give η_{n-1} and η_n , respectively. Since $\eta_0 = C_0 = 1$, all these norms equal 1. Changing from monomials to monic orthogonal polynomials is unitriangular, and therefore $H_N^{(0)} = \prod_{n=0}^{N-1} \eta_n = 1$.

Next set $z = 0$ in (3.5) and use (3.3). We obtain

$$\frac{H_N^{(m+1)}}{H_N^{(m)}} = (-1)^N p_N^{(m)}(0) = \frac{4^N (m + \frac{1}{2})_N}{(N + m + 1)_N} = \prod_{j=1}^m \frac{2N + m + j}{m + j}. \quad (3.9)$$

For completeness, the middle expression equals

$$\frac{(2m + 2N)! m! (N + m)!}{(m + N)! (2m)! (2N + m)!} = \frac{(2N + 2m)! m!}{(2m)! (2N + m)!},$$

which is the last product. Induction on m , beginning with $H_N^{(0)} = 1$, proves (1.5).

3.4 Completion of the coefficient proof

Insert (3.3) into (3.6). The constant factor is $H_N^{(m+1)}$ by (3.9). In each summand, $(-N)_k (-1)^k / k! = \binom{N}{k}$. This gives exactly (2.1). The formula is a polynomial identity for $b \neq 0$, so it holds for $b = 0$ as well. The duplication identity for a rising half-integer factorial, or direct cancellation, gives (2.2). Taking adjacent coefficient ratios gives (2.3).

All factors in (2.1) are positive rational numbers. The left side is a determinant with entries in $\mathbb{Z}[a, b]$, so its coefficients are integers. This proves positivity and integrality simultaneously and completes Theorem 2.1. In particular,

$$c_{N,m,0} = H_N^{(m+1)}, \quad c_{N,m,N} = H_N^{(m)}. \quad (3.10)$$

3.5 Real roots and coefficient inequalities

Corollary 3.3. *For $N \geq 1$, the polynomial $a \mapsto D_N^{(m)}(a, 1)$ has exactly N simple real roots, all in $(-4, 0)$. For $N \geq 2$, its normalized coefficients $u_k = c_{N,m,k} / \binom{N}{k}$ satisfy*

$$u_k^2 > u_{k-1} u_{k+1} \quad (1 \leq k < N). \quad (3.11)$$

Proof. A degree- N orthogonal polynomial for a strictly positive density on $(0, 4)$ has N simple roots in that interval. One elementary proof is to multiply it by the product of its distinct sign-changing roots in $(0, 4)$. If there were fewer than N such roots, that product would have degree below N , while the integral of its product with the orthogonal polynomial would have constant nonzero sign. This contradicts orthogonality. Equation (3.6) now negates the roots.

Alternatively, the coefficient inequality follows without any root theory. Theorem 2.1 gives

$$\frac{u_{k+1}}{u_k} = \frac{N + m + 1 + k}{4(m + \frac{1}{2} + k)}.$$

This is strictly decreasing in k , because the numerator's constant term exceeds the denominator's unscaled constant term by $N + 1/2$. The decrease is equivalent to (3.11). $\square$

4 A fixed-size confluent determinant

The explicit coefficient formula grows in length with N . To understand the recurrence order, we need a representation whose determinant size depends only on m . Define

$$R_n(y) = (-1)^n p_n^{(0)}(-y) = \sum_{i=0}^n \binom{n+i}{2i} y^i. \quad (4.1)$$

Thus

$$R_0(y) = 1, \quad R_1(y) = y + 1, \quad R_{n+2}(y) = (y + 2)R_{n+1}(y) - R_n(y). \quad (4.2)$$

Extend R_n to all integer indices using this recurrence. Since $R_{-1} = R_0 = 1$, it also satisfies the reflection identity

$$R_{-n-1}(y) = R_n(y) \quad (n \in \mathbb{Z}). \quad (4.3)$$

The reversed form of the polynomials is convenient because it eliminates unnecessary signs from Christoffel's identity.

Lemma 4.1 (Christoffel determinant in the present normalization). *For pairwise distinct $y_1, \dots, y_r$,*

$$\det_{0 \leq i, j < N} \int_0^4 x^{i+j} \prod_{\ell=1}^r (x + y_\ell) d\mu_0(x) = \frac{\det[R_{N+j}(y_i)]_{\substack{1 \leq i \leq r \\ 0 \leq j < r}}}{\prod_{1 \leq i < j \leq r} (y_j - y_i)}. \quad (4.4)$$

The quotient extends polynomially when some y_i coincide.

Proof. Here is a proof that also verifies the normalization. Write $\Delta(x_1, \dots, x_N) = \det[x_i^{j-1}]_{1 \leq i, j \leq N}$. Expanding two determinants and integrating termwise gives the identity

$$\frac{1}{N!} \int \cdots \int \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu_0(x_j) = \det \left[\int f_i(x) g_j(x) d\mu_0(x) \right].$$

No analytic interchange difficulty occurs here: the sums are finite and all polynomials are integrable on a bounded interval.

For external variables $z_1, \dots, z_r$ we have

$$\Delta(x, z) = \Delta(x) \Delta(z) \prod_{i=1}^N \prod_{\ell=1}^r (z_\ell - x_i).$$

In the determinant defining $\Delta(x, z)$, replace monomials by the monic polynomials $p_j^{(0)}$; the determinant is unchanged. Expand along its first N rows and integrate against $\Delta(x)$. By orthogonality, every choice of columns vanishes except columns of degrees $0, \dots, N-1$. That surviving integral equals $N! H_N^{(0)} = N!$. The remaining minor is $\det[p_{N+j}^{(0)}(z_i)]$. Dividing by $\Delta(z)$ yields the identity for the multiplier $\prod_\ell (z_\ell - x)$.

Finally put $z_\ell = -y_\ell$ and use $p_{N+j}^{(0)}(-y) = (-1)^{N+j} R_{N+j}(y)$. The signs from the multiplier, column factors, and Vandermonde cancel, giving (4.4). The numerator is alternating in the y_i , hence divisible by their Vandermonde as a polynomial. This proves the extension statement. This is the classical fixed-size identity discussed in [3]; its inclusion here is a normalization proof, not a novelty claim. $\square$

Take $r = m + 1$, let the first m variables tend to zero, and put the last one equal to $y = a/b \neq 0$. Dividing by the Vandermonde and taking the limit replaces those first rows by Taylor coefficients. Since $[y^i] R_{N+j}(y) = \binom{N+j+i}{2i}$, we obtain:

Proposition 4.2 (Confluent representation). *If $ab \neq 0$, then*

$$D_N^{(m)}(a, b) = b^N y^{-m} \det \begin{pmatrix} \binom{N+j+i}{2i} & (0 \leq i < m, 0 \leq j \leq m) \\ R_{N+j}(y) & (0 \leq j \leq m) \end{pmatrix}, \quad y = a/b. \quad (4.5)$$

The upper block has m rows and the lower block has one row. For $m = 0$ the formula simply reads $D_N^{(0)} = b^N R_N(a/b)$.

Define the $m \times m$ minors

$$M_{m,\ell}(N) = \det_{\substack{0 \leq i < m \\ j \in J_\ell}} \binom{N+j+i}{2i}, \quad J_\ell = \{0, 1, \dots, m\} \setminus \{\ell\}, \quad (4.6)$$

where columns are in increasing order and $0 \leq \ell \leq m$. Expansion of (4.5) along its last row gives

$$D_N^{(m)}(a, b) = b^N y^{-m} \sum_{\ell=0}^m (-1)^{m+\ell} M_{m,\ell}(N) R_{N+\ell}(y). \quad (4.7)$$

Although a naive degree bound for $M_{m,\ell}$ is $m(m-1) = 2d$, its actual degree is only d . This is the key point.

5 The degree cancellation and its first correction

Lemma 5.1 (Sharp minor degree). *For $0 \leq \ell \leq m$, $M_{m,\ell}(N)$ is a polynomial of degree $d = \binom{m}{2}$ with leading coefficient*

$$[N^d]M_{m,\ell}(N) = h_m \binom{m}{\ell}. \quad (5.1)$$

For $d \geq 1$, the next coefficient is

$$[N^{d-1}]M_{m,\ell}(N) = h_m \binom{m}{\ell} \left(\frac{d(m+2)}{2} - \frac{(m-1)\ell}{2} \right). \quad (5.2)$$

Proof. Let $f_i(x) = \binom{x+i}{2i}$, so f_i has degree $2i$ and leading coefficient $1/(2i)!$. The determinant $\det[f_i(x_j)]_{0 \leq i, j < m}$ is alternating in its m independent variables. It is divisible by $\Delta(x) = \prod_{p < q} (x_q - x_p)$, whose degree is d . The determinant has total degree at most $\sum_i 2i = 2d$; its quotient by Δ therefore has total degree at most d . Substituting $x_j = N + j$ for $j \in J_\ell$ makes Δ a nonzero constant. Hence the degree in N is at most d .

To obtain the leading coefficient, expand $f_i(N + j)$ in Taylor powers of j . The first possible nonzero determinant uses distinct Taylor orders $0, 1, \dots, m-1$; these have the least possible sum d . The coefficient of N^d is

$$\frac{\det[\binom{2i}{r}]_{0 \leq i, r < m}}{\prod_{i=0}^{m-1} (2i)!} \Delta(J_\ell) = \frac{2^d \Delta(J_\ell)}{\prod_{i=0}^{m-1} (2i)!}. \quad (5.3)$$

To see the second equality, regard $\binom{2i}{r}$ as a polynomial of degree r in i with leading coefficient $2^r/r!$ and evaluate its determinant at $i = 0, \dots, m-1$. The Vandermonde is $\prod_{r=1}^{m-1} r!$, so these factorials cancel. Removing ℓ from $0, \dots, m$ also gives

$$\Delta(J_\ell) = \binom{m}{\ell} \prod_{r=1}^{m-1} r!. \quad (5.4)$$

Since

$$h_m = \frac{2^d \prod_{r=1}^{m-1} r!}{\prod_{i=0}^{m-1} (2i)!} \quad (m \geq 1),$$

we obtain (5.1). It is nonzero, proving the degree assertion. The case $m = 0$ is the empty determinant and is immediate.

For the next coefficient, observe the symmetric factorization

$$f_i(x) = \frac{1}{(2i)!} \prod_{r=1}^i \left(\left(x + \frac{1}{2}\right)^2 - \left(r - \frac{1}{2}\right)^2 \right). \quad (5.5)$$

Thus $f_i(x) = (x + \frac{1}{2})^{2i} / (2i)!$ plus terms of degree at most $2i - 2$. Replacing one or more rows by those lower-degree terms reduces the total degree by at least two. The same alternating argument then bounds their degree after translation by $d - 2$. The leading two terms are consequently obtained from

$$\det \left[\frac{(N + j + \frac{1}{2})^{2i}}{(2i)!} \right]_{0 \leq i < m, j \in J_\ell} = \frac{\prod_{\substack{p < q \\ p, q \in J_\ell}} (q - p)(2N + p + q + 1)}{\prod_{i=0}^{m-1} (2i)!}. \quad (5.6)$$

The relative coefficient of N^{d-1} is

$$\frac{1}{2} \sum_{\substack{p < q \\ p, q \in J_\ell}} (p + q + 1) = \frac{1}{2} \left((m-1) \left(\frac{m(m+1)}{2} - \ell \right) + d \right),$$

which simplifies to (5.2). If $d = 1$, the putative remainder of degree $d - 2 = -1$ is identically zero. $\square$

It is useful to collect the minors in a polynomial in an auxiliary variable r :

$$Q_m(N; r) = \sum_{\ell=0}^m (-1)^{m+\ell} M_{m,\ell}(N) r^\ell. \quad (5.7)$$

By the binomial theorem,

$$[N^d] Q_m(N; r) = h_m(r - 1)^m. \quad (5.8)$$

When $r \neq 1$ and $d \geq 1$, the relative next coefficient is

$$\frac{[N^{d-1}] Q_m(N; r)}{[N^d] Q_m(N; r)} = \frac{d(m+2)}{2} - d \frac{r}{r-1} = d \left(\frac{m}{2} - \frac{1}{r-1} \right). \quad (5.9)$$

Indeed, applying $r \, d/dr$ to $(r - 1)^m$ sums the terms weighted by ℓ in (5.2). The exact value d , rather than an upper bound $2d$, is what makes the denominator exponent sharp.

6 Two exponentials and the minimal recurrence

Suppose first that $b \neq 0$ and $a(a + 4b) \neq 0$. Set $y = a/b$ and choose one ordering of the two roots

$$r_{\pm} = \frac{y + 2 \pm \sqrt{y(y+4)}}{2}, \quad r_+ r_- = 1. \quad (6.1)$$

They are distinct and nonzero. Solving the recurrence (4.2) with its two initial values gives

$$R_n(y) = \frac{(r_+ - 1)r_+^n - (r_- - 1)r_-^n}{r_+ - r_-}. \quad (6.2)$$

Write

$$\lambda_{\pm} = b r_{\pm} = \frac{a + 2b \pm \sqrt{a(a+4b)}}{2}. \quad (6.3)$$

Substitution in (4.7) gives the exact identity

$$D_N^{(m)}(a, b) = \frac{(r_+ - 1)Q_m(N; r_+)\lambda_+^N - (r_- - 1)Q_m(N; r_-)\lambda_-^N}{y^m(r_+ - r_-)}. \quad (6.4)$$

In particular, both polynomial coefficients in this two-exponential representation have degree *exactly* d : neither r_+ nor r_- can equal 1, since $y \neq 0$.

Lemma 6.1 (Minimal annihilator of an exponential polynomial). *If $\lambda_1, \dots, \lambda_s$ are distinct nonzero complex numbers and the nonzero polynomials $A_i(N)$ have degrees d_i , then $u_N = \sum_i A_i(N)\lambda_i^N$ has minimal constant-coefficient annihilator*

$$\prod_{i=1}^s (E - \lambda_i)^{d_i+1}, \quad Eu_N = u_{N+1}. \quad (6.5)$$

Its generating function has poles of orders $d_i + 1$ at $t = 1/\lambda_i$.

Proof. On $A(N)\lambda^N$, the operator $E - \lambda$ is $\lambda^{N+1}(A(N+1) - A(N))$; it lowers the polynomial degree by exactly one until the result vanishes. For $\mu \neq \lambda$, $E - \mu$ preserves that degree, multiplying the leading coefficient by $\lambda - \mu$. The generalized eigenspaces for distinct eigenvalues are independent: polynomial projectors from the Chinese remainder identity isolate each one. Thus every annihilator needs the stated multiplicity at every eigenvalue. This argument remains valid after any finite shift in N .

Equivalently, applying $(z d/dz)^k$ to $1/(1-z)$ shows that $\sum_{N \geq 0} N^k z^N$ has a pole of exact order $k+1$ at $z=1$, with nonzero leading pole coefficient $k!$. Distinct pole locations cannot cancel. This proves the generating-function assertion too. $\square$

Applying the lemma to (6.4) proves the minimal characteristic polynomial (2.6). It also proves that $q^{d+1}F_m$ is a polynomial of degree at most $2d+1 = 2e-1$ for these parameters. Section 7 sharpens its degree to K .

6.1 Why the identity extends to all parameters

Each $D_N^{(m)}$ is a polynomial in a, b with integer coefficients. The coefficients of $q^{d+1}F_m$ are therefore such polynomials too. Every coefficient that vanishes for all $b \neq 0$ and $a(a+4b) \neq 0$ vanishes identically: a polynomial cannot be nonzero and vanish on that nonempty open set. Hence the uniform rational identity, and its coefficient integrality, extend to every a, b , even where its displayed denominator is not reduced. This argument is coefficientwise in a formal series and does not require a convergence limit in t .

6.2 Proof of the exceptional classification

When $b = 0$,

$$D_N^{(m)}(a, 0) = a^N H_N^{(m)}. \quad (6.6)$$

For $a \neq 0$, the polynomial $H_N^{(m)}$ has degree d , so Lemma 6.1 gives denominator $(1-at)^{d+1}$. When $a = 0$ and $b \neq 0$,

$$D_N^{(m)}(0, b) = b^N H_N^{(m+1)}, \quad (6.7)$$

whose polynomial degree is $\binom{m+1}{2}$.

At the remaining collision $a = -4b$, with $b \neq 0$, the recurrence gives

$$R_n(-4) = (-1)^n (2n+1). \quad (6.8)$$

Inserting this into (4.7) and tracking the signs produces

$$D_N^{(m)}(-4b, b) = (-b)^N E_m(N), \quad (6.9)$$

$$E_m(N) = 4^{-m} \sum_{\ell=0}^m (2N + 2\ell + 1) M_{m,\ell}(N). \quad (6.10)$$

By (5.1), its degree is exactly $d + 1$, with

$$[N^{d+1}]E_m(N) = 4^{-m} 2h_m \sum_{\ell=0}^m \binom{m}{\ell} = 2^{1-m} h_m \neq 0. \quad (6.11)$$

Thus the reduced denominator is $(1 + bt)^{d+2}$. Finally, if $a = b = 0$, every nonempty determinant vanishes and $F_m = 1$. Together with the nonexceptional result, this proves Theorem 2.3.

The two root-collision lines have different pole orders. This is not a contradiction: $x = 0$ and $x = 4$ are inequivalent endpoints of the weight $x^{m-1/2}(4-x)^{1/2}$. In particular, blindly substituting $a = 0$ into a generic minimal-order statement would be incorrect.

7 The numerator degree and reciprocal symmetry

A rational function can have a denominator of large degree and a surprisingly short numerator. Here the explanation is a string of zero values in the natural bilateral extension of the determinant.

For the moment retain the nonexceptional hypotheses. Define D_N for all $N \in \mathbb{Z}$ by the right side of (4.5). The upper rows are interpreted through their polynomial binomial coefficients; this agrees with the Taylor coefficients of R_n at zero, including negative n , because

$$\binom{-n-1+i}{2i} = \binom{n+i}{2i}.$$

It also agrees with the unique bilateral extension supplied by (6.4).

Lemma 7.1 (Reflection and a zero interval). *Let $\sigma_m = (-1)^{\binom{m+1}{2}}$. Then*

$$D_{-N-m-1} = \sigma_m b^{-2N-m-1} D_N \quad (N \in \mathbb{Z}), \quad (7.1)$$

$$D_{-1} = D_{-2} = \cdots = D_{-m} = 0. \quad (7.2)$$

The second statement is empty for $m = 0$.

Proof. The index change $N \mapsto -N - m - 1$, followed by $R_{-n-1} = R_n$, reverses all $m + 1$ columns of the determinant. The sign of that reversal is σ_m . Comparing its prefactor b^N on the two sides gives (7.1). If $N = -s$ with $1 \leq s \leq m$, columns $j = s - 1$ and $j = s$ correspond to R_{-1} and R_0 , respectively, which are the same polynomial. Their values and all needed Taylor coefficients coincide, so the determinant vanishes. $\square$

Lemma 7.2 (The expansion at infinity). *For a sequence of exponential-polynomial form with nonzero characteristic roots and its bilateral extension, its proper rational generating function satisfies*

$$F(t) = - \sum_{r \geq 1} D_{-r} t^{-r} \quad (7.3)$$

as a formal Laurent expansion at infinity.

Proof. For $D_N = \lambda^N$ the assertion is simply $(1 - \lambda t)^{-1} = - \sum_{r \geq 1} \lambda^{-r} t^{-r}$. Apply powers of $t \, d/dt$ to obtain the assertion for $D_N = N^k \lambda^N$, and then use linearity. $\square$

Combining (7.2), (7.1), and $D_0 = 1$ yields

$$F_m(t) = -\sigma_m b^{-m-1} t^{-m-1} + O(t^{-m-2}) \quad (t \rightarrow \infty). \quad (7.4)$$

Since q^e has degree $2e$ and leading coefficient b^{2e} , the numerator has degree

$$2e - m - 1 = m(m - 1) + 2 - m - 1 = (m - 1)^2 = K \quad (7.5)$$

and leading coefficient $-\sigma_m b^{2e-m-1} = s_m b^K$. The identity $s_m = -\sigma_m$ follows because the difference of the two exponents $\binom{m+1}{2}$ and $(m - 1)(m - 2)/2$ is $2m - 1$, an odd integer. This calculation proves the sharp degree assertion, including $m = 0$.

The full reflection identity yields more than the leading term. Putting $1/(b^2 t)$ into (7.3) and reindexing with (7.1), we find

$$F_m\left(\frac{1}{b^2 t}\right) = s_m b^{m+1} t^{m+1} F_m(t). \quad (7.6)$$

But $q(1/(b^2 t)) = q(t)/(b^2 t^2)$. Substitution into (7.6) gives exactly (2.5). After denominators in b are cleared, each of these numerator identities is polynomial in a, b . Thus they extend to the exceptional parameters; for $b \neq 0$ the leading coefficient remains nonzero. This completes the proof of Theorem 2.2.

7.1 An explicit finite prescription for the numerator

No uncomputed infinite series is needed to determine P_m . Expand

$$q(t)^e = \sum_{j=0}^{2e} q_j t^j. \quad (7.7)$$

For example, its coefficients have the finite expression

$$q_j = \sum_{v=\max(0, j-e)}^{\lfloor j/2 \rfloor} \frac{e!}{(j-2v)! v! (e-j+v)!} (-(a+2b))^{j-2v} b^{2v}. \quad (7.8)$$

Then

$$P_m(t) = \sum_{\ell=0}^K \left(\sum_{j=0}^{\min(\ell, 2e)} q_j D_{\ell-j}^{(m)}(a, b) \right) t^\ell, \quad (7.9)$$

where every determinant in the right side is given by Theorem 2.1. Equations (7.8) and (7.9) therefore constitute a finite sum/product formula for every numerator coefficient. All coefficients of $q^e F_m$ after degree K vanish.

The usual recurrence is

$$\sum_{j=0}^{2e} q_j D_{N-j}^{(m)}(a, b) = 0 \quad (N \geq 2e). \quad (7.10)$$

In fact the truncated convolution already vanishes for $N > K$ when negative-index terms are taken as zero for the purpose of multiplying ordinary power series. Those artificial zeros should not be confused with the distinct bilateral extension used in the proof above.

8 Exact exponential expansions and asymptotics

Equation (6.4) supplies every coefficient of the two polynomials in Theorem 2.4. Explicitly,

$$B_+(N) = \frac{(r_+ - 1)Q_m(N; r_+)}{y^m(r_+ - r_-)}, \quad B_-(N) = -\frac{(r_- - 1)Q_m(N; r_-)}{y^m(r_+ - r_-)}. \quad (8.1)$$

These formulas are valid for arbitrary nonexceptional complex parameters; the assumption $a, b > 0$ is only needed to select a strictly dominant real exponential for the asymptotic statement.

For $a, b > 0$, write $r = r_+$, so $r_- = r^{-1}$ and

$$y = r + r^{-1} - 2 = \frac{(r-1)^2}{r}. \quad (8.2)$$

Using (5.8), the coefficient of N^d in B_+ is

$$\frac{h_m(r-1)^{m+1}}{y^m(r-r^{-1})} = h_m \frac{r^{m+1}}{(r-1)^m(r+1)} = A_m.$$

Its relative next coefficient is exactly (5.9). Also $\lambda_-/\lambda_+ = r^{-2}$. Division of (6.4) by $A_m N^d \lambda_+^N$ proves (2.9).

One may write the complete answer in the finite form

$$D_N^{(m)} = \lambda_+^N \sum_{j=0}^d \beta_j^+ N^{d-j} + \lambda_-^N \sum_{j=0}^d \beta_j^- N^{d-j}, \quad (8.3)$$

where

$$\beta_j^+ = \frac{r_+ - 1}{y^m(r_+ - r_-)} [N^{d-j}] Q_m(N; r_+), \quad \beta_j^- = -\frac{r_- - 1}{y^m(r_+ - r_-)} [N^{d-j}] Q_m(N; r_-). \quad (8.4)$$

Thus the “full asymptotic expansion” is particularly strong here: it is an exact expression, with no uncomputed remainder once both finite sectors are retained. No assertion of asymptotic convergence or summability is needed. The leading coefficient ratio of the two sectors is

$$\frac{\beta_0^-}{\beta_0^+} = (-1)^m r^{-(m+1)}. \quad (8.5)$$

These estimates are for fixed m, a, b . They are not claimed to be uniform when $a \rightarrow 0$, when the two roots coalesce, or when m grows with N . In such regimes the exact finite formula remains valid away from the collision itself, but a different uniform asymptotic analysis would be required. At a collision, use the exact exceptional formulas from Section 6 instead.

9 Worked examples and integer sequences

9.1 Small shifts

The first uniform generating functions are

$$F_0(t) = \frac{1-bt}{q(t)}, \quad F_1(t) = \frac{1}{q(t)}, \quad (9.1)$$

$$F_2(t) = \frac{1+bt}{q(t)^2}, \quad F_3(t) = \frac{(1-b^2t^2)(1+(a+6b)t+b^2t^2)}{q(t)^4}. \quad (9.2)$$

They can be obtained directly from (7.9). Notice that $m = 0$ is a genuine boundary case: its numerator has degree 1, not 0. For $m = 3$, the numerator is anti-reciprocal; for $m = 2$, it is reciprocal. This is the pattern described uniformly by s_m .

For $m = 2$, put $A = a + 2b$. The recurrence becomes

$$D_N = 2AD_{N-1} - (A^2 + 2b^2)D_{N-2} + 2Ab^2D_{N-3} - b^4D_{N-4} \quad (N \geq 4). \quad (9.3)$$

The initial polynomials are

$$\begin{aligned} D_0 &= 1, \\ D_1 &= 2a + 5b, \\ D_2 &= 3a^2 + 14ab + 14b^2, \\ D_3 &= 4a^3 + 27a^2b + 54ab^2 + 30b^3. \end{aligned}$$

The recurrence is minimal outside the exceptional lines; displaying only a valid order-four relation would not by itself prove that fact.

9.2 The specialization $a = b = 1$

The first sequences, all beginning with the empty determinant, are

$m \backslash N$	0	1	2	3	4	5	6
0	1	2	5	13	34	89	233
1	1	3	8	21	55	144	377
2	1	7	31	115	390	1254	3893
3	1	19	170	1075	5580	25529	107036

The first two are F_{2N+1} and F_{2N+2} , respectively, as follows from their common recurrence $u_{N+2} = 3u_{N+1} - u_N$ and initial values. They connect the family to the odd and even Fibonacci subsequences in OEIS A001519 and A001906 [6]; Catalan numbers themselves are A000108 [5]. The higher rows and their coefficient triangles are natural OEIS-style objects, but the archive does not assert that any particular row is absent from OEIS.

For every fixed shift, the dominant exponential is

$$\lambda_+ = \frac{3 + \sqrt{5}}{2},$$

while its polynomial prefactor has degree $\binom{m}{2}$. Thus increasing the shift changes the power of N , not the exponential growth constant. For $m = 2$, the first relative asymptotic correction equals $1 - 1/(r - 1) = (3 - \sqrt{5})/2$ with $r = (3 + \sqrt{5})/2$.

9.3 Rational characteristic roots: $a = 1, b = 2$

This specialization gives $r_+ = 2, r_- = 1/2$ and $\lambda_+ = 4, \lambda_- = 1$. Therefore the exact formulas have rational polynomial coefficients and no radicals:

$$D_N^{(0)}(1, 2) = \frac{2 \cdot 4^N + 1}{3}, \quad (9.4)$$

$$D_N^{(1)}(1, 2) = \frac{4 \cdot 4^N - 1}{3}, \quad (9.5)$$

$$D_N^{(2)}(1, 2) = \frac{8N 4^N + N + 3}{3}, \quad (9.6)$$

$$D_N^{(3)}(1, 2) = \frac{(32N^3 + 48N^2 + 16N + 96)4^N - (2N^3 + 21N^2 + 73N + 78)}{18}. \quad (9.7)$$

For example, the $m = 2$ sequence starts 1, 12, 87, 514, 2733, 13656, ... Its dominant polynomial is $(8/3)N$ with no constant term. This vanishing agrees with the first-correction formula $d(m/2 - 1/(r - 1)) = 1(1 - 1) = 0$; the entire correction is the smaller exponential sector $(N + 3)/3$. For $m = 3$, the relative coefficient of N^{-1} in the dominant sector is $3/2$, again exactly as predicted.

9.4 An endpoint example

For $m = 2$, (6.10) simplifies to

$$E_2(N) = \frac{(N+1)(N+2)}{2}.$$

Hence

$$D_N^{(2)}(-4b, b) = (-b)^N \frac{(N+1)(N+2)}{2}, \quad F_2(t; -4b, b) = \frac{1}{(1+bt)^3}. \quad (9.8)$$

This also follows by cancelling $1+bt$ from $(1+bt)/(1+bt)^4$, the uniform form of F_2 . At the other collision, $a = 0$, one instead obtains $F_2(t; 0, b) = (1+bt)/(1-bt)^4$. These explicit cases illustrate why the reduced denominator needs a separate classification.

10 Algorithms, exact experiments, and reproducibility

10.1 Implementation choices

The file `code/catalan_hankel.py` implements the coefficient formula, pure shifted product, homogeneous polynomial evaluation, fixed-size determinant, and uniform and reduced denominators. All numerical parameters in its public computation interface are integers; internal fractions use Python's exact `Fraction`. The mathematical theorems have the broader complex-parameter scope stated above. No floating-point arithmetic enters the tests.

For a coefficient row, first compute the triangular product $H_N^{(m+1)}$ and then apply (2.3) successively. The number of arithmetic operations is $O(m^2 + N)$; this is an *arithmetic-operation count*, not a bit-complexity claim. Every division in the update is checked for exactness. Homogeneous Horner evaluation avoids division by a or b , so zero parameters are handled without special numerical limits.

The independent determinant path uses a fraction-free Bareiss algorithm on the original matrix of Catalan moments, including pivoting when a diagonal pivot vanishes. This matters for signed parameters: an implementation tested only on positive definite examples can conceal pivot failures. The source-indexing formula is implemented separately, not obtained by calling the corrected coefficient routine.

For many consecutive values at fixed m , it is also possible to use (7.10) after producing the first $2e$ values. For very large isolated indices, powering the companion matrix or reducing X^N modulo (2.6) provides standard recurrence-based methods. These methods are consequences of the theorem; the supplied reference implementation prioritizes transparent exact verification rather than optimized large-integer throughput.

10.2 Verification that was actually run

The included `data/verification.json` records PASS for all checks in the following table. The report is generated by the test program, not manually written.

Check	Count
Direct determinant versus coefficient evaluation	1,183
Additional direct signed/degenerate parameter evaluations	810
Printed source formula versus corrected shift	990
Confluent fixed-size determinant evaluations	364
Minor polynomials exactly interpolated and checked	36
Generating-function/recurrence coefficient identities	2,816
Numerator reciprocal coefficient identities	604
Nonexceptional numerator–denominator gcd checks	44
Exceptional reduced-denominator checks	44
Exact two-exponential evaluations	168
Strict normalized log-concavity inequalities	5,655

The first 1,183 evaluations cover every pair $0 \leq N, m \leq 12$, using $a = 0, 1, \dots, N$ and $b = 1$. At each fixed (N, m) , both sides are polynomials of degree at most N in a . Equality at $N + 1$ distinct points certifies that particular polynomial identity exactly. Thus this part certifies 169 full coefficient polynomials, not merely 169 evaluations. It still does not prove the result for untested values of N or m ; the proof in Section 3 does that.

For $0 \leq m \leq 7$, the minor tests interpolate using $2d + 2$ points, more than enough for the elementary degree bound $2d$. They verify the drop to degree d , both leading coefficients from Lemma 5.1, and additional negative-index values. The recurrence checks include $0 \leq m \leq 10$ and four nonexceptional parameter pairs, while separate tests examine every exceptional class. The exact two-exponential tests use $(a, b) = (1, 2)$, $m \leq 7$, and $0 \leq N \leq 20$. These tests check different parts of the derivation, but are not represented as formal proof certificates for arbitrary parameters.

10.3 How to reproduce the package

From the archive's top directory, run:

```
python3 code/verify.py --out data
python3 code/catalan_hankel.py 8 3 --a 2 --b 3 --direct
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Use Python 3.10 or later and do not pass -0 , because the verification suite deliberately uses assertions. The script writes coefficient rows, generating functions, minor polynomials, exact exponential-sector polynomials, and integer sequence tables in JSON or CSV. Rational polynomial coefficients are stored as strings such as "8/3", in ascending power order, to avoid any loss of exactness when imported into other systems. The included README documents every file and its indexing convention. [Added 28 September 2026, batch 39: the first command regenerates the shipped files in data/; in the repository, run it on a copy (see the report README). Part II's separate suite, and why its delivered commands must not be run in this directory, are described in Section 27.]

11 What has been resolved, and what has not

The all-shift denominator formula is proved here, with a stronger minimality assertion and a complete description of the degenerate parameters. The coefficient expression is likewise proved for all N, m ; comparison with the exact formula printed in the target source requires the explicit indexing correction in the next appendix. The reciprocal numerator and finite exponential expansions are proved from the same structural determinant calculation.

These are substantive mathematical conclusions about the selected conjectural family. They are not evidence that the Christoffel identity, Jacobi representation, Catalan shifted product, or general rationality

are new. A literature search was performed against the source title, arXiv identifier, denominator phrase, and related recurrence work. The cited version still labels the sharp exponent conjectural, and the retrieved sources did not supply an explicit statement of this full sharp classification. That is a bounded search result, not a proof that no earlier equivalent theorem exists. Before a priority claim or submission, the exact formulas should be checked against the wider orthogonal-polynomial and determinant literature and by a human referee.

Several natural extensions remain separate from the theorem. One may seek a direct combinatorial interpretation of the individual positive integers $c_{N,m,k}$, or a product formula for the minors $M_{m,\ell}(N)$ beyond their first two coefficients. Multiple distinct linear factors in the multiplier lead to more exponential sectors and new collision patterns. A joint limit with growing m or a parameter approaching $a = 0$ would require uniform, rather than fixed-parameter, asymptotics. None of these extensions is assumed in the present proofs.

[Added 28 September 2026, batch 39: the preceding paragraph describes the report at the state fbba58593 against which Part II was written. The extension to several distinct linear factors, and to repeated factors, is treated in Part II: for every nonzero polynomial multiplier the exponential sectors, their exact degrees and nonzero leading coefficients (Theorems 14.2 and 18.1), the minimal order off resonance (14.12), an exact detector of the parameters where it drops (Theorem 14.5), and a certified finite algorithm on that resonance locus (Theorems 20.1 and 20.2). Part II gives no short closed classification of the resonant cancellations; it poses that as a research direction (Section 24). The other extensions named here remain open: Part II does not address a combinatorial interpretation of $c_{N,m,k}$, a product formula for the minors $M_{m,\ell}(N)$, or joint limits with growing m ; and its asymptotics (Section 22) are for fixed parameters, so a parameter approaching $a = 0$ (a root reaching the endpoint 0) is among the uniform endpoint limits it poses, not solves.]

Part II. Root collisions: arbitrary polynomial multipliers

Sections 1–11 and Appendices A–B form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition, a line on the title page, contents entries for the two parts, and two bibliography entries used only by Part II. Part II was added on 28 September 2026 in batch 39 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the extension that Part I’s Section 11 named but left open, “multiple distinct linear factors in the multiplier” and their “new collision patterns”: for every nonzero polynomial multiplier it determines the exponential sectors, their exact polynomial degrees and leading coefficients, the minimal recurrence off resonance, and a finite algorithm on the resonance locus. Section 12 records the provenance and fixes the notation of the addition.

12 Provenance and conventions of Part II

12.1 One manuscript, one addition

Part II is batch-39 manuscript 03, *Root Collisions and Sharp Recurrences for Polynomially Weighted Catalan Hankel Determinants*, a proof-focused research article prepared with OpenAI ChatGPT for Vladimir Reshetnikov’s ProveIt research program and dated 28 September 2026 (23 pages as delivered). It arrived as the archive `ProveIt_Catalan_Root_Collisions.zip` (inner directory `ProveIt_Catalan_Root_Collisions/`, main file `article.tex`) in ProveIt commit `74f7f5bdb` and was placed in `e2b1f016a`. It was written against ProveIt commit `fbba58593dc0622aa914972896150d4848f935b5`, reading Part I’s `article.tex` and `README.md` at that commit (consulted 28 September 2026 through the pinned repository tree). Part I’s files are unchanged between that commit and the placement commit, so every statement the manuscript makes about “the repository report” refers to Part I as printed above.

The manuscript’s code and data are shipped beside Part I’s files with the prefix `02-root-collisions-` (Part I’s unprefix files count as 01), and its provenance notes, delivered as `notes/provenance.md`, as `02-root-collisions-provenance.md` at the report root. Its manuscript, delivery README and PDF are not shipped; they survive in the arrival commit. Section 27 lists the shipped files.

The manuscript contributes, in the order printed here: its research question and a statement of what is classical (Section 13); the definitions and principal results, namely the exact sector-degree law, the minimal recurrence off resonance and the secondary-Hankel factorization (Section 14); the orthogonal polynomials and the confluent Christoffel identity in its normalization (Section 15); the local jets at interior points and at both endpoints (Section 16); the determinant degree calculus (Section 17); the canonical sectors and the exact leading coefficient (Section 18); minimality, the order formula and the secondary factorization (Section 19); resonance, finite-prefix certification and a root-free algorithm (Section 20); worked examples, including the recovery of Part I’s theorem (Section 21); exact asymptotics (Section 22); its verification record, implementation interface and a formalization route (Section 23); eight research directions (Section 24); a conclusion; a normalization and sign audit (Section 26); and its reproduction instructions (Section 27). Every result, proof, example, remark, limitation and question of the manuscript is printed. Where it re-derives something Part I already proves, the overlap is marked in the text (Section 12.4).

The manuscript’s own abstract reads:

For a nonzero polynomial q , let $H_N(q)$ be the size- N Hankel determinant of the Catalan moments multiplied by q . General constant-coefficient recurrences for this family are classical. We determine their sharp multiplicities. Write $q(x) = \gamma x^m (x-4)^\ell \prod_i (x-c_i)^{t_i}$, with distinct $c_i \notin \{0, 4\}$, and put $c_i = 2 + z_i + z_i^{-1}$. The branch-labelled exponential with rate $(-1)^{\ell+\sum t_i} \gamma \prod_i z_i^{2k_i-t_i}$ has a polynomial

coefficient whose degree is exactly $\binom{m}{2} + \binom{\ell+1}{2} + \sum_i k_i(t_i - k_i)$. A nonvanishing confluent-Vandermonde product gives its leading coefficient. Consequently, when the rates are distinct, the minimal recurrence order is $\prod_i (t_i + 1) \left(\binom{m}{2} + \binom{\ell+1}{2} + 1 + \frac{1}{6} \sum_i t_i(t_i - 1) \right)$. We prove an exact secondary-Hankel factorization that detects every loss of this maximal order, and give a finite, root-free algorithm determining the minimal recurrence at all rational or algebraic specializations. For one repeated nonendpoint root the generic order is cubic in the multiplicity, rather than the classical exponential bound. The example $q = (2x - 9)^4$ has order 15, while resonant fourth powers can have order 5. The proofs extend the repository's linear-multiplier result to arbitrary root-multiplicity patterns, including both endpoints, and are accompanied by exact-arithmetic verification code.

Status (manuscript 03). This is an unrefereed, AI-assisted mathematical manuscript. The universal results are proved in the text; the finite computations are an independent audit, not a substitute for those arguments. No Lean formalization or claim of first discovery is made. General polynomial-multiplier rationality, the Christoffel formula and its confluent form are classical and are not presented as new; the proposed contribution is the exact branch degree, its nonvanishing leading constant, the resulting minimality classification off resonance, and their assembled structural and algorithmic consequences. No novelty is claimed for the secondary-Hankel identity of an arbitrary exponential polynomial (Lemma 19.2), an elementary confluent-Vandermonde calculation. The literature comparison was targeted (the sources cited in Section 13) and does not establish that the theorem has never appeared in another notation; Schur-function, Toeplitz/Hankel determinant and orthogonal-polynomial literature may contain equivalent specializations, and expert review and a broader priority search would be needed before any first-discovery claim. No current global open-problem status is asserted for anything, including the eight research directions of Section 24. The theorems are over characteristic zero and are not to be transported unchanged to small positive characteristic. Nothing in Part II is formalized in any proof assistant, and its placement beside a Lean project confers no formal status. (Sources: the manuscript and its provenance notes, 02-root-collisions-provenance.md.)

12.2 Reading conventions

Inside Sections 13–27, “this article” and “the present article” mean manuscript 03 as printed in Part II, “the repository report” and “the ProveIt report” mean Part I at the pin fbba58593, and “the archive” or “the package” means the delivered package, whose code and data are shipped under the prefixed names listed in Section 27. The manuscript’s two appendices are printed as Sections 26 and 27, so that Part I’s appendices keep their letters. The manuscript cites Krattenthaler’s Ramanujan Journal paper in its preprint form arXiv:2101.04225v5 (10 September 2021), and points to its Theorem 1, Proposition 5 and Section 8, Corollary 9; that is the same paper as Part I’s reference [3]. The manuscript cites Barry’s note [1] for the denominator-exponent assertion in its introduction and does not reassert the printed coefficient conjecture without the index correction of Appendix A.

12.3 Notation of Part II

Part II uses the manuscript’s symbols, which differ from Part I’s in many places. Within Sections 13–27 every symbol has its Part II meaning, except where a sentence names Part I or cites one of its equations. Part I’s parameters a, b appear only where Part I’s multiplier $x^m(a + bx)$ is discussed: the first subsection of Section 13 and the first subsection of Section 21. Elsewhere a and b are local letters (roots, indices, block counts, a leading coefficient). Table 1 lists the symbols that a reader of Part I is likely to misread. One symbol was renamed: the manuscript’s maximal order R_0 is written $\mathcal{R}_0$, because Part I’s $R_0(y) = 1$ is the first of its polynomials (4.1). No normalization changed.

Table 1: Symbols of Part II. The last column gives Part I's meaning of the same letter, or the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
$q(x)$; subscripts in $\Delta_q, \chi_q, \Psi_q, B_q$	the nonzero multiplier polynomial (14.2)	Part I's $q(t) = 1 - (a + 2b)t + b^2t^2$ (1.4), the quadratic denominator; Part II's generating-function variable is u , not t .
$H_N(q)$	the weighted Hankel determinant (14.1)	Part I's $H_N^{(m)} = H_N(x^m)$ and $D_N^{(m)}(a, b) = H_N(x^m(a + bx))$ are special cases; in Theorem 20.2, $h = (h_N)$ is an arbitrary sequence, not Part I's h_m .
m	multiplicity of the root 0 of q	agrees with Part I's shift m when $q = x^m(a + bx)$.
ℓ	multiplicity of the root 4 of q	Part I's column index in $M_{m,\ell}$ and Q_m .
$s; s(z), s_i$	number of distinct roots outside $\{0, 4\}$; the scalar $1/(z - z^{-1})$ of (16.2)	Part I's sign s_m in (2.5), and the source shift s of Appendix A.
c_i, z_i, t_i	roots outside $\{0, 4\}$, spectral parameters $c_i = 2 + z_i + z_i^{-1}$, multiplicities	Part I's coefficients $c_{N,m,k}$ and its variable t ; for $q = x^m(a + bx)$, $-z^{\pm 1}$ are Part I's $r_{\pm}$.
T, D	$\sum_i t_i$ and $\deg q$	Part I's $T_{n,k,s}$ (Appendix A) and its determinant $D_N^{(m)}$.
$\mathcal{K}, \mathbf{k}, \kappa$	index set (14.4), a label in it, and $\binom{m}{2} + \binom{\ell+1}{2}$	Part I's numerator degree $K = (m - 1)^2$.
λ_k	rate of the sector $\mathbf{k}$ (14.6)	for $q = x^m(a + bx)$ the two rates are Part I's $\lambda_{\pm}$ (6.3).
d_k, r_k	exact sector degree (14.7) and $d_k + 1$	Part I's $d = \binom{m}{2}$ and $e = d + 1$ (the linear case), its roots $r_{\pm}$ and the variable r of $Q_m(N; r)$.
$\mathcal{R}_0$ (renamed from R_0)	the maximal recurrence order (14.12)	Part I's polynomial $R_0(y) = 1$ (4.1); Part II's plain R is another order or size (an assumed bound in Theorem 20.2, a matrix size in Lemma 19.2, the root-free bound (20.6)).
$P_k(N), L_k$	sector polynomial in the size N (18.4), and its leading coefficient	Part I's numerator $P_m(t; a, b)$, a polynomial in t ; Part I's sector polynomials are $B_{\pm}(N)$.
$A_{\pm}(z), A_{r,\pm}(n; z)$	branch amplitudes and jet polynomials (16.4)	Part I's asymptotic constant A_m (2.10).
$B_0(m), B_4(\ell), B_z(k); B_i; B_q(u)$	local leading determinants (18.7), (17.5); blocks in Lemma 19.2; the numerator in (19.3)	Part I's $B_{\pm}(N)$ of Theorem 2.4.
$M_N; M$	the $D \times D$ Christoffel matrix (15.7); $\max_{\mathbf{k}} d_{\mathbf{k}}$ in Section 22	Part I's minors $M_{m,\ell}(N)$ (4.6).
E	forward shift, $(Ef)(N) = f(N + 1)$	Part I's shift E in Lemma 6.1 and its endpoint polynomial $E_m(N)$ (6.10).
Λ, Q_{λ}	set of distinct rates; grouped sector polynomial (20.1)	Part I's minor generating polynomial $Q_m(N; r)$ (5.7).
$(a_u, v_u)_{u \leq b}; (w_a, u_a)_{a \leq b}; b_t$	the distinct roots of q , their multiplicities and number b (Section 15); nodes and block sizes in Lemma 17.1; degree of the square-free factor f_t	Part I's parameters a, b , which Part II uses only in the first subsections of Sections 13 and 21.
p_n, μ	monic orthogonal polynomials (15.1) and the Catalan measure	they are Part I's $p_n^{(0)}$ and μ_0 ; Part I's $p_N^{(m)}$ and μ_m with $m \geq 1$ are different.

Symbol	Meaning in Part II	Not to be confused with
$\Delta_q, V, \mathcal{V}$	cross-root product (15.6); Vandermonde (matrix); confluent product (18.6)	Part I's Vandermonde $\Delta(x)$.
C (proof of Theorem 18.1)	a matrix of coefficient polynomials	the Catalan numbers C_n .

12.4 Where the merge had to choose

1. *Placement.* Part II is printed after Part I's Section 11 and before Part I's appendices, so that no section, equation, theorem or table number of Part I moves (checked against the `.aux` file of a build of the committed text). Part II's sections continue the numbering (Sections 12 to 27); the manuscript's Sections 1–13 are Sections 13–25, and its Appendices A and B are Sections 26 and 27.
2. *Labels.* Every label of Part II carries the prefix `rc::`; without it, the manuscript's `sec:main`, `thm:main`, `eq:R`, `eq:lambda`, `eq:measure`, `lem:Christoffel`, `eq:Christoffel` and several others would have collided with Part I's.
3. *Overlaps with Part I.* The manuscript re-derives, without assuming Part I correct, several things Part I already proves. Printed with a pointer and kept: the orthogonality of the p_n and $\det(C_{i+j}) = 1$ (a second route; Section 15); the Christoffel identity (Part I's Lemma 4.1 is its special case $\gamma = 1$, with Part I's normalization R_n , and Proposition 4.2 its confluent case at the root 0); the binomial determinant (17.3) (as in (5.3)); the minimal-annihilator lemma (Lemma 19.1 restates Lemma 6.1; the manuscript's proof is kept as a second proof); and the recovery of Part I's sharp order and exceptional orders (a second proof by a different route, Section 21). No statement of Part I was changed.
4. *Bibliography.* The manuscript's references to Barry and to Krattenthaler's Ramanujan Journal paper are Part I's [1, 3]; its versions and pointers are recorded in Section 12.2. Its references to Cigler and Krattenthaler and to Krattenthaler's Uvarov-formula preprint are added to the common bibliography. Its reference to the pinned repository report is replaced by Part I itself and the pin recorded in Section 12.1.
5. *File names.* Sections 23 and 27 give the shipped names beside the delivered ones. The manuscript's reproduction commands are printed as delivered but must not be run in place here; Section 27 gives a recipe on a scratch copy.
6. *Front matter.* The manuscript's title page, abstract, status paragraph and archive paragraph are replaced by Section 12.1, which prints the abstract and the status. The archive paragraph is not reprinted because the delivered source and PDF are not shipped; the shipped code and data are listed in Section 27.
7. *Notation.* The manuscript's symbols are kept, with the single rename $R_0 \rightarrow \mathcal{R}_0$ and the scoping convention and Table 1 of Section 12.3.

13 The research question and its provenance

13.1 A concrete extension of the ProveIt report

The starting point is the report *Shifted Catalan Hankel Polynomials: Exact coefficients, sharp recurrences, and a denominator conjecture*, dated 19 September 2026, in ProveIt, printed above as Part I (§12.1 records the pinned snapshot). For

$$D_N^{(m)}(a, b) = \det_{0 \leq i, j < N} (aC_{i+j+m} + bC_{i+j+m+1}), \quad C_n = \frac{1}{n+1} \binom{2n}{n},$$

it proves the sharp characteristic polynomial

$$(X^2 - (a + 2b)X + b^2)^{\binom{m}{2}+1} \quad (13.1)$$

when $b \neq 0$ and $a(a + 4b) \neq 0$ (this is (2.6) of Theorem 2.2). Its concluding discussion (Section 11) explicitly leaves multiple linear factors and their collision patterns for further work. That is the extension addressed here.

The repository was inspected at commit

fbba58593dc0622aa914972896150d4848f935b5.

The relevant source is under

SetTheory/Cardinals/docs/reports/hankel-determinants/
catalan-and-ballot/shifted-catalan-hankel-polynomials/article.tex.

The present proof does not assume that an unrefereed repository manuscript is correct. We rederive the determinant identity and the local degree calculus, then recover (13.1) as a specialization.

The question is not merely whether another rational generating function exists. It is this:

Given every root and its multiplicity in a polynomial multiplier, which exponential terms actually occur, what is the exact polynomial degree of each term, and precisely when does the minimal recurrence order drop?

A repeated root is not equivalent to several independently chosen roots. The associated determinant forces cancellations within a repeated block. Those cancellations are the main subject of this article.

13.2 What is classical, and what is being added

The Christoffel determinant formula and its confluent form are classical; Krattenthaler's treatment [3] gives several proofs and explicitly handles coalescing evaluation points. Its Section 8, Corollary 9, supplies a constant-coefficient recurrence of order at most 2^D for a multiplier of degree D , including Catalan moments. That account also credits Elouafi's earlier proof of the Catalan recurrence assertion. General rationality is therefore not a new conclusion of this article.

Barry's note [1] motivates the shifted linear combination and its quadratic-denominator exponent. The earlier work of Cigler and Krattenthaler [7] gives further context for polynomial linear combinations of moments. Our comparison is with these particular sources and the pinned repository report, not with an exhaustive classification of the literature.

The additions proved below are an exact degree law for every repeated-root block, its nonzero leading constant with endpoint factors included, the resulting sharp recurrence and a factorization detecting its exceptional locus. A root-free finite procedure handles resonances without assuming that a generic formula remains minimal after specialization. The underlying determinant techniques are familiar; the full multiplicity law and its assembled consequences are the proposed research contribution. An equivalent earlier formulation in the wider orthogonal-polynomial or symmetric-function literature has not been ruled out.

14 Definitions and principal results

14.1 The weighted determinant and its spectral parameters

For $q(x) = \sum_{r=0}^D q_r x^r \in \mathbb{C}[x] \setminus \{0\}$, define

$$H_N(q) = \det_{0 \leq i, j < N} \left(\sum_{r=0}^D q_r C_{i+j+r} \right), \quad N \in \mathbb{N}_0, \quad H_0(q) = 1. \quad (14.1)$$

Write its factorization as

$$q(x) = \gamma x^m (x - 4)^\ell \prod_{i=1}^s (x - c_i)^{t_i}, \quad (14.2)$$

where $\gamma \neq 0$, $m, \ell \in \mathbb{N}_0$, $t_i \geq 1$, and the c_i are pairwise distinct and avoid 0, 4. Thus

$$T = \sum_{i=1}^s t_i, \quad D = m + \ell + T.$$

An empty product is 1. Constant multipliers are included by taking $m = \ell = s = 0$. The identically zero multiplier is excluded; its sequence would simply be 1, 0, 0, ...

For each i choose a solution of

$$c_i = 2 + z_i + z_i^{-1}. \quad (14.3)$$

Then $z_i \in \mathbb{C}^\times \setminus \{1, -1\}$. Replacing z_i by z_i^{-1} only relabels an index below. No final recurrence depends on this choice. Define

$$\mathcal{K} = \prod_{i=1}^s \{0, 1, \dots, t_i\}, \quad (14.4)$$

$$\kappa = \binom{m}{2} + \binom{\ell + 1}{2}, \quad (14.5)$$

$$\lambda_{\mathbf{k}} = (-1)^{\ell+T} \gamma \prod_{i=1}^s z_i^{2k_i - t_i}, \quad (14.6)$$

$$d_{\mathbf{k}} = \kappa + \sum_{i=1}^s k_i(t_i - k_i), \quad r_{\mathbf{k}} = d_{\mathbf{k}} + 1. \quad (14.7)$$

When $s = 0$, the index set $\mathcal{K}$ has one element, the empty tuple.

Definition 14.1 (Resonance). The parameters are *nonresonant* if the values $\lambda_{\mathbf{k}}$, $\mathbf{k} \in \mathcal{K}$, are pairwise distinct. Otherwise there is a *resonance*. Equivalently, a resonance is an identity

$$\prod_i z_i^{2(k_i - j_i)} = 1 \quad \text{for some distinct } \mathbf{k}, \mathbf{j} \in \mathcal{K}. \quad (14.8)$$

This is different from a repeated root of q : repeated roots have already been recorded by the t_i .

14.2 The exact multiplicity law

Theorem 14.2 (Branch degrees and minimal recurrence). *For every multiplier (14.2), there are canonically branch-labelled polynomials $P_{\mathbf{k}} \in \mathbb{C}[N]$ such that*

$$H_N(q) = \sum_{\mathbf{k} \in \mathcal{K}} P_{\mathbf{k}}(N) \lambda_{\mathbf{k}}^N \quad (N \geq 0), \quad (14.9)$$

and

$$\deg P_{\mathbf{k}} = d_{\mathbf{k}} \quad \text{for every } \mathbf{k}. \quad (14.10)$$

Their leading coefficients are given by the nonzero product in Theorem 18.1. These individual assertions remain valid at resonances, when different branch labels can have the same numerical rate.

If the parameters are nonresonant, the minimal characteristic polynomial of the sequence is

$$\chi_q(X) = \prod_{\mathbf{k} \in \mathcal{K}} (X - \lambda_{\mathbf{k}})^{d_{\mathbf{k}} + 1}. \quad (14.11)$$

Its degree is

$$\mathcal{R}_0 = \prod_{i=1}^s (t_i + 1) \left(\kappa + 1 + \frac{1}{6} \sum_{i=1}^s t_i(t_i - 1) \right). \quad (14.12)$$

For arbitrary parameters, (14.11) is still an annihilating polynomial of degree $\mathcal{R}_0$. The true minimal order equals $\mathcal{R}_0$ if and only if the parameters are nonresonant.

A characteristic polynomial $g(X) = \sum_{j=0}^r g_j X^j$, with $g_r = 1$, means the recurrence

$$\sum_{j=0}^r g_j H_{N+j}(q) = 0 \quad \text{for every } N \geq 0. \quad (14.13)$$

Thus the assertions concern the entire sequence, including its initial values, not only an eventual recurrence. A polynomial of minimal degree with this property is unique.

Corollary 14.3 (One repeated nonendpoint root). *Let $q(x) = \gamma(x-c)^t$, where $c \notin \{0, 4\}$ and $c = 2+z+z^{-1}$. If $z^{2h} \neq 1$ for $1 \leq h \leq t$, then*

$$\text{ord}(H(q)) = t + 1 + \binom{t+1}{3}. \quad (14.14)$$

The exponential with rate $(-1)^t \gamma z^{2k-t}$ has polynomial degree $k(t-k)$. In particular the first orders, for $t = 1, \dots, 8$, are

$$2, 4, 8, 15, 26, 42, 64, 93.$$

The order in (14.14) is asymptotic to $t^3/6$, in contrast to the universal bound 2^t . The first strict improvement occurs at $t = 4$. This is a uniform family of exact minimality results, rather than an isolated fitted recurrence.

Corollary 14.4 (Simple factors and endpoints). *If every $t_i = 1$ and the parameters are nonresonant, the minimal order is*

$$2^s \left(\binom{m}{2} + \binom{\ell+1}{2} + 1 \right). \quad (14.15)$$

If $s = 0$, no nonresonance hypothesis is needed: $H_N(q)$ is $((-1)^\ell \gamma)^N$ times a polynomial of degree exactly κ . The minimal polynomial is $(X - (-1)^\ell \gamma)^{\kappa+1}$.

The asymmetry between 0 and 4 is essential. A multiplicity m at 0 contributes $\binom{m}{2}$ degrees, whereas a multiplicity ℓ at 4 contributes $\binom{\ell+1}{2}$. Treating both endpoints as a generic collision of two equally weighted branches gives the wrong answer.

14.3 An exact detector for exceptional parameters

Put $L_k = \text{lc}(P_k)$, and choose any ordering of $\mathcal{K}$.

Theorem 14.5 (Secondary Hankel factorization). *With $\mathcal{R}_0 = \sum_k r_k$, one has*

$$\begin{aligned} & \det_{0 \leq u, v < \mathcal{R}_0} (H_{u+v}(q)) \\ &= (-1)^{\sum_k \binom{r_k}{2}} \prod_k \left((d_k! L_k)^{r_k} \lambda_k^{r_k(r_k-1)} \right) \prod_{k < j} (\lambda_j - \lambda_k)^{2r_k r_j}. \end{aligned} \quad (14.16)$$

This identity holds at resonant as well as nonresonant parameters. All factors outside the final product are nonzero. Consequently this secondary determinant vanishes exactly on the resonance locus.

The formula is stronger than a statement that a generic matrix has full rank. It identifies every source of rank loss and its multiplicity in spectral coordinates. Sections 19 and 20 prove this assertion and make it algorithmic.

15 Catalan orthogonality and the fixed-size determinant

This section proves the classical infrastructure with our normalization. It is included to make the signs, initial index and confluent factorials fully auditable. The general identities are not claimed as new.

15.1 The orthogonal polynomials

Define monic polynomials by

$$p_0(x) = 1, \quad p_1(x) = x - 1, \quad p_{n+1}(x) = (x - 2)p_n(x) - p_{n-1}(x) \quad (n \geq 1). \quad (15.1)$$

The Catalan probability measure is

$$d\mu(x) = \frac{1}{2\pi} \sqrt{\frac{4-x}{x}} \mathbf{1}_{(0,4)}(x) dx. \quad (15.2)$$

Substitution $x = 4u$ and the beta integral give

$$\int x^n d\mu(x) = \frac{2 \cdot 4^n}{\pi} B\left(n + \frac{1}{2}, \frac{3}{2}\right) = \frac{(2n)!}{n!(n+1)!} = C_n.$$

There is also an elementary trigonometric verification of the orthogonality. For $x = 2 + 2 \cos \theta$, $0 < \theta < \pi$, the recurrence gives

$$p_n(x) = \frac{\sin((n + \frac{1}{2})\theta)}{\sin(\theta/2)}, \quad d\mu = \frac{1 - \cos \theta}{\pi} d\theta. \quad (15.3)$$

The product-to-sum formula then yields

$$\int p_n(x) p_j(x) d\mu(x) = \delta_{n,j}. \quad (15.4)$$

Indeed, after cancellation of $\sin^2(\theta/2)$ the integral is $\frac{2}{\pi} \int_0^\pi \sin((n + \frac{1}{2})\theta) \sin((j + \frac{1}{2})\theta) d\theta$. The change from monomials to monic p_n is triangular with determinant 1. Hence (15.4) also proves

$$\det_{0 \leq i, j < N} (C_{i+j}) = 1. \quad (15.5)$$

Relation to Part I. The measure μ is Part I's μ_0 of (3.1), and p_n is Part I's $p_n^{(0)}$, with the same recurrence (3.8). Part I proves (15.5), $H_N^{(0)} = 1$, from the constant squared norms of the $p_n^{(0)}$ (Section 3); the trigonometric orthogonality above is a second route, kept as the manuscript printed it.

15.2 Christoffel's determinant, including confluence

List the distinct roots in (14.2) in the order $0, 4, c_1, \dots, c_s$, omitting endpoints whose multiplicities are zero. Write this list as $(a_1, \nu_1), \dots, (a_b, \nu_b)$ and set

$$\Delta_q = \prod_{1 \leq u < v \leq b} (a_v - a_u)^{\nu_u \nu_v}. \quad (15.6)$$

It is nonzero. Let M_N have rows indexed by (a_u, r) , $0 \leq r < \nu_u$, and columns indexed by $j = 0, \dots, D-1$, with entries

$$(M_N)_{(a_u, r), j} = \frac{p_{N+j}^{(r)}(a_u)}{r!}. \quad (15.7)$$

Lemma 15.1 (Confluent Christoffel identity). *For all $N \geq 0$,*

$$H_N(q) = (-1)^{ND} \gamma^N \frac{\det M_N}{\Delta_q}. \quad (15.8)$$

Proof. First suppose all roots $a_1, \dots, a_D$ are distinct. For an ordered list y , write $V(y) = \prod_{i < j} (y_j - y_i)$. Expanding two determinants under the integral and using the product measure proves the elementary integration identity

$$\int \det(f_i(x_j))_{i,j=1}^N \det(g_i(x_j))_{i,j=1}^N \prod_j d\mu(x_j) = N! \det \left(\int f_i(x) g_j(x) d\mu(x) \right)_{i,j=1}^N.$$

Taking monomials with the multiplier included gives

$$H_N(q) = \frac{1}{N!} \int V(x_1, \dots, x_N)^2 \prod_{i=1}^N q(x_i) d\mu(x_i).$$

Since

$$V(x, a) = V(x)V(a) \prod_{i,u} (a_u - x_i),$$

we obtain

$$H_N(q) = \frac{(-1)^{ND} \gamma^N}{N! V(a)} \int V(x) V(x, a) \prod_i d\mu(x_i). \quad (15.9)$$

Replace the monomials in both Vandermonde determinants by the monic polynomials p_j . Expand $V(x, a)$ along its first N rows. On integration against $V(x)$, orthogonality kills every selected set of columns except $0, 1, \dots, N-1$. That surviving integral is $N!$, and its complementary minor is $(p_{N+j}(a_u))_{u,j}$. The Laplace sign is positive because the selected columns are the first N . This proves (15.8) for distinct roots.

Now let ν distinct roots tend to a common value a . Taylor expansion shows that division by their internal Vandermonde replaces their successive evaluation rows by the rows $p_{N+j}^{(r)}(a)/r!$, $r = 0, \dots, \nu-1$. Cross-block differences tend to the product (15.6). Applying this separately to every block gives the stated formula. Equivalently, both sides are polynomial identities in the root variables after the removable Vandermonde factors are canceled. Thus the limiting argument does not assume that any weighted Hankel determinant is nonzero. At $N = 0$, both the empty moment determinant and the normalized confluent Vandermonde equal 1. $\square$

The important computational consequence is that the determinant on the right has size D , independent of N . The important theoretical consequence is different: all dependence on N has been localized in jets of a two-branch orthogonal-polynomial sequence.

Relation to Part I. Lemma 4.1 of Part I is the case $\gamma = 1$, roots $a_u = -y_u$, in the reversed normalization $R_n(y) = (-1)^n p_n(-y)$ of (4.1); its proof follows the same integration-and-Laplace route for distinct roots. Part I then takes the confluent limit only for the m -fold root at 0 (Proposition 4.2). Lemma 15.1 adds confluence at every root, both endpoints, and the leading coefficient γ with its sign, so it is printed in full.

16 Local jet structure

16.1 The two branches away from the endpoints

The characteristic equation of (15.1), at $x = 2 + z + z^{-1}$, has roots z, z^{-1} . Matching the first two values gives

$$p_n(2 + z + z^{-1}) = \frac{z^{n+1} - z^{-n}}{z - 1} = A_+(z)z^n + A_-(z)z^{-n}, \quad (16.1)$$

where

$$A_+(z) = \frac{z}{z-1}, \quad A_-(z) = -\frac{1}{z-1}, \quad s(z) = \frac{1}{z-z^{-1}}. \quad (16.2)$$

The letter $s(z)$ is a scalar function, not the number s of distinct nonendpoint roots. For clarity we often write $s_i = s(z_i)$. The inverse function theorem applies locally because $z \neq \pm 1$, and

$$\frac{d}{dx} \log z(x) = s(z), \quad \frac{d}{dx} = z s(z) \frac{d}{dz}. \quad (16.3)$$

Lemma 16.1 (Offendpoint jets). *For $c = 2 + z + z^{-1} \notin \{0, 4\}$ and $r \geq 0$,*

$$\frac{p_n^{(r)}(c)}{r!} = z^n A_{r,+}(n; z) + z^{-n} A_{r,-}(n; z), \quad (16.4)$$

where $A_{r,\pm}(n; z)$ are polynomials of degree exactly r in n . Their leading coefficients are

$$[n^r] A_{r,\pm}(n; z) = \frac{A_{\pm}(z)(\pm s(z))^r}{r!}. \quad (16.5)$$

They can be constructed recursively from $A_{0,\pm} = A_{\pm}$ by

$$A_{r+1,\pm}(n; z) = \frac{zs(z) \partial_z A_{r,\pm}(n; z) \pm ns(z) A_{r,\pm}(n; z)}{r+1}. \quad (16.6)$$

Proof. Differentiate (16.4) once and use (16.3). The normalization by $(r+1)!$ gives (16.6). The derivative term does not increase the degree in n , while multiplication by $\pm ns(z)$ increases it by one. Induction gives (16.5). All its factors are nonzero, so the degree is exact. $\square$

16.2 The two endpoint jets

From (15.1),

$$\sum_{n \geq 0} p_n(x) u^n = \frac{1+u}{1-(x-2)u+u^2}. \quad (16.7)$$

Differentiation in x , normalized by $r!$, gives

$$\sum_{n \geq 0} \frac{p_n^{(r)}(x)}{r!} u^n = \frac{(1+u)u^r}{(1-(x-2)u+u^2)^{r+1}}. \quad (16.8)$$

Lemma 16.2 (Endpoint degrees). *For all $n, r \geq 0$,*

$$\frac{p_n^{(r)}(0)}{r!} = (-1)^{n-r} \binom{n+r}{2r}, \quad (16.9)$$

$$\frac{p_n^{(r)}(4)}{r!} = \frac{2n+1}{2r+1} \binom{n+r}{2r}. \quad (16.10)$$

After the common factor $(-1)^n$ is removed at 0, the polynomial in n has degree $2r$ and leading coefficient $(-1)^r/(2r)!$. At 4 the degree is $2r+1$, with leading coefficient $2/(2r+1)!$.

Proof. At $x = 0$, (16.8) becomes $u^r/(1+u)^{2r+1}$, which gives (16.9). At $x = 4$ it becomes $(1+u)u^r/(1-u)^{2r+2}$. The coefficient is

$$\binom{n+r+1}{2r+1} + \binom{n+r}{2r+1} = \frac{2n+1}{2r+1} \binom{n+r}{2r}.$$

The assertions on leading coefficients follow immediately. For $n < r$ the binomial coefficient is zero, as it must be for a derivative whose order exceeds the polynomial degree. $\square$

The same quadratic recurrence has two different endpoint amplitudes: $p_n(0) = (-1)^n$ but $p_n(4) = 2n+1$. This single extra degree at 4 is what ultimately changes $\binom{\ell}{2}$ into $\binom{\ell+1}{2}$.

17 The determinant degree calculus

17.1 Polynomial shifts and confluent exponential rows

The next two identities isolate the cancellation responsible for the sharp order. If f is a polynomial of degree d with leading coefficient a , then

$$[j^h]f(N+j) \text{ has degree at most } d-h \text{ in } N, \quad [N^{d-h}j^h]f(N+j) = a \binom{d}{h}. \quad (17.1)$$

An expression with $h > d$ is interpreted as zero.

For distinct nonzero numbers $w_1, \dots, w_b$ and positive integers $u_1, \dots, u_b$ summing to D , form the matrix

$$V_{(a,h),j} = w_a^j j^h, \quad 0 \leq h < u_a, \quad 0 \leq j < D.$$

The convention $0^0 = 1$ is used only in the polynomial value j^0 .

Lemma 17.1 (Confluent exponential Vandermonde). *In the indicated block order,*

$$\det V = \prod_{a=1}^b \left(w_a^{\binom{u_a}{2}} \prod_{h=0}^{u_a-1} h! \right) \prod_{a < b} (w_b - w_a)^{u_a u_b}. \quad (17.2)$$

Proof. Start with the ordinary Vandermonde determinant and coalesce u_a variables to each w_a . The matrix of ordinary, unnormalized derivatives of w^j then has determinant $\prod_a \prod_{h < u_a} h! \prod_{a < b} (w_b - w_a)^{u_a u_b}$. Within a block replace ∂_w^h by $(w \partial_w)^h$. This is a triangular change of rows: the coefficient of ∂_w^h is w^h , and all other terms have smaller derivative order. The determinant is multiplied by $\prod_a w_a^{0+1+\dots+(u_a-1)}$. Since $(w \partial_w)^h w^j = j^h w^j$, the resulting matrix is V . $\square$

17.2 Three local leading determinants

For an endpoint block of size u , the leading matrices obtained from (17.1) have entries proportional to $\binom{2r}{h}$ or $\binom{2r+1}{h}$, $0 \leq r, h < u$. As a polynomial in its upper argument, $\binom{x}{h}$ has leading coefficient $1/h!$. Consequently

$$\det_{0 \leq r, h < u} \binom{2r}{h} = \det_{0 \leq r, h < u} \binom{2r+1}{h} = 2^{\binom{u}{2}}. \quad (17.3)$$

Here the Vandermonde in $0, 2, \dots, 2u-2$ or $1, 3, \dots, 2u-1$ contributes $2^{\binom{u}{2}} \prod_{h < u} h!$, canceling the column leading denominators. (The first equality with $u = m$ is the determinant evaluated in (5.3) of Part I, by the same argument.)

For the nonendpoint calculation let $t = k + h$, and write $A_+ = A_+(z)$, $A_- = A_-(z)$ and $s = s(z)$. Consider the $t \times t$ matrix with rows $r = 0, \dots, t-1$, followed first by k plus columns and then by h minus columns:

$$B_{r,(+,j)} = \frac{A_+ s^r}{r!} \binom{r}{j} \quad (0 \leq j < k), \quad B_{r,(-,j)} = \frac{A_- (-s)^r}{r!} \binom{r}{j} \quad (0 \leq j < h). \quad (17.4)$$

Lemma 17.2 (The mixed-block determinant). *The determinant in (17.4) equals*

$$B_z(k) = \frac{A_+^k A_-^h s^{\binom{k}{2}} (-s)^{\binom{h}{2}} (-2s)^{kh}}{\prod_{r=0}^{t-1} r!}, \quad h = t - k, \quad (17.5)$$

and is nonzero.

Proof. Remove the row factors $1/r!$ and the column factors $A_{\pm}$. Use

$$w^r \binom{r}{j} = w^j \frac{1}{j!} \left(\frac{d}{dw} \right)^j w^r.$$

The remaining normalized-derivative matrix is the transpose of a confluent Vandermonde at $w = s$ and $w = -s$. Its determinant is $(-2s)^{kh}$. The column powers supply $s^{\binom{k}{2}}(-s)^{\binom{h}{2}}$, proving (17.5). Every factor is nonzero because $z \notin \{0, 1, -1\}$. $\square$

This lemma is the point at which a mere degree bound becomes an exact degree theorem. It sums all competing assignments of the derivative rows to the two branches, including their determinant signs. Ignoring those signs, or arguing that one assignment has a nonzero leading term, would not establish noncancellation of the whole sector.

18 Canonical sectors and the proof of sharpness

18.1 A definition that still makes sense at resonance

Set

$$f_{0,r}(n) = (-1)^r \binom{n+r}{2r}, \quad f_{4,r}(n) = \frac{2n+1}{2r+1} \binom{n+r}{2r}.$$

Introduce independent marker variables $y_1, \dots, y_s$. Let $\tilde{M}(N; \mathbf{y})$ be the $D \times D$ matrix whose column j contains the following rows, in the same root order as before:

$$(-1)^j f_{0,r}(N+j), \quad 0 \leq r < m, \quad (18.1)$$

$$f_{4,r}(N+j), \quad 0 \leq r < \ell, \quad (18.2)$$

$$y_i z_i^j A_{r,+}(N+j; z_i) + z_i^{-j} A_{r,-}(N+j; z_i), \quad 0 \leq r < t_i. \quad (18.3)$$

Its determinant is a polynomial in N and the markers. Define

$$P_{\mathbf{k}}(N) = \frac{1}{\Delta_q} [y_1^{k_1} \cdots y_s^{k_s}] \det \tilde{M}(N; \mathbf{y}). \quad (18.4)$$

This definition does not require distinct numerical rates. It is the meaning of “canonically branch-labelled” in Theorem 14.2.

In the original matrix M_N , remove $(-1)^N$ from each of its m zero-endpoint rows and z_i^{-N} from each of the t_i rows at c_i . The remaining matrix is (18.1)–(18.3) with $y_i = z_i^{2N}$. Therefore (15.8) gives

$$H_N(q) = \left((-1)^{D+m} \gamma \prod_i z_i^{-t_i} \right)^N \sum_{\mathbf{k}} P_{\mathbf{k}}(N) \prod_i z_i^{2N k_i}.$$

Since $D+m$ has the parity of $\ell + T$, this proves (14.9), with the rates in (14.6).

18.2 The leading coefficient in product form

For each label $\mathbf{k}$, form an ordered list of spectral blocks

$$\mathcal{W}_{\mathbf{k}} = ((-1, m), (1, \ell), (z_1, k_1), (z_1^{-1}, t_1 - k_1), \dots, (z_s, k_s), (z_s^{-1}, t_s - k_s)), \quad (18.5)$$

omitting all blocks of size zero. Its nodes are pairwise distinct: an equality between a node from c_i and a node from c_j would imply $c_i = c_j$, and none is an endpoint node ± 1 . For any such ordered list write

$$\mathcal{V}((w_a, u_a)_{a=1}^b) = \prod_a \left(w_a^{\binom{u_a}{2}} \prod_{h=0}^{u_a-1} h! \right) \prod_{a < b} (w_b - w_a)^{u_a u_b}. \quad (18.6)$$

Finally define

$$B_0(m) = \frac{(-2)^{\binom{m}{2}}}{\prod_{r=0}^{m-1} (2r)!}, \quad B_4(\ell) = \frac{2^{\binom{\ell+1}{2}}}{\prod_{r=0}^{\ell-1} (2r+1)!}. \quad (18.7)$$

Both equal 1 at multiplicity zero.

Theorem 18.1 (Exact leading coefficient). *For P_k defined by (18.4),*

$$[N^{d_k}]P_k(N) = \frac{B_0(m)B_4(\ell) \prod_{i=1}^s B_{z_i}(k_i)}{\Delta_q} \mathcal{V}(\mathcal{W}_k) \neq 0. \quad (18.8)$$

Here $B_{z_i}(k_i)$ is (17.5) with $t = t_i$. Moreover P_k has no term of degree greater than d_k .

Proof. Expand each row of $\tilde{M}$ in the finite collection of basis rows

$$(w^j j^h)_{j=0}^{D-1}. \quad (18.9)$$

Only the finitely many h up to the largest relevant polynomial degree are needed. This factors $\tilde{M} = C(N; \mathbf{y})V$, with C a matrix of coefficient polynomials and V the matrix of basis rows. Cauchy–Binet expresses its determinant as a sum over selections of D basis rows.

A nonzero coefficient minor must select exactly m rows associated with node -1 , exactly ℓ with node 1 , and exactly t_i rows from the two nodes z_i, z_i^{-1} for each i . Otherwise a root block of C , whose support is confined to its own nodes, cannot have full rank. The coefficient of $\mathbf{y}^k$ selects precisely k_i rows at z_i and $t_i - k_i$ rows at z_i^{-1} .

For a fixed node with u selected basis rows, their h indices must be distinct. Their sum is therefore at least $0 + 1 + \dots + (u-1) = \binom{u}{2}$, with equality only for the set $\{0, \dots, u-1\}$. By (17.1), selecting such an index h subtracts h from the maximum possible degree in N of its coefficient. Thus the zero-endpoint block contributes at most

$$\sum_{r=0}^{m-1} 2r - \binom{m}{2} = \binom{m}{2},$$

the four-endpoint block contributes at most

$$\sum_{r=0}^{\ell-1} (2r+1) - \binom{\ell}{2} = \binom{\ell+1}{2},$$

and the i th nonendpoint block contributes at most

$$\sum_{r=0}^{t_i-1} r - \binom{k_i}{2} - \binom{t_i - k_i}{2} = k_i(t_i - k_i).$$

The sum of these bounds is d_k .

There is exactly one selection of basis rows that can attain this total degree: at every node take $h = 0, \dots, u-1$. The associated basis determinant is (18.6) for the list $\mathcal{W}_k$, by Lemma 17.1. In its coefficient minor, the root blocks separate. For the endpoint 0, the leading row factors from Lemma 16.2, followed by (17.3), give $B_0(m)$. At 4 they give $B_4(\ell)$. At c_i the leading coefficient matrix is exactly (17.4), so Lemma 17.2 gives $B_{z_i}(k_i)$. Dividing by Δ_q proves (18.8).

Every factor in (18.8) is nonzero. In particular, the unique maximal-degree basis selection cannot vanish or be canceled by another selection, because every other selection has strictly smaller degree. This proves both exactness and noncancellation. $\square$

Remark 18.2 (Why the naive degree is too large). For a block of multiplicity t , multiplying row degrees without taking the determinant into account suggests $\binom{t}{2}$ degrees. The k plus-branch rows consume $\binom{k}{2}$ degrees, and the $t - k$ minus rows consume $\binom{t-k}{2}$. The surviving degree is their difference, $k(t - k)$. The consumption is forced by linear dependence of equal low-degree basis rows; it is not an accidental cancellation at special numerical parameters.

19 Minimal recurrences, order, and the secondary factorization

19.1 Exponential polynomials and minimality

Let $(Ef)(N) = f(N + 1)$ denote forward shift.

Lemma 19.1 (Minimal annihilators). *Suppose $\lambda_1, \dots, \lambda_b$ are distinct nonzero complex numbers and Q_i are nonzero polynomials of degrees d_i . Then $f(N) = \sum_i Q_i(N)\lambda_i^N$ has minimal characteristic polynomial*

$$\prod_{i=1}^b (X - \lambda_i)^{d_i+1}. \quad (19.1)$$

Its ordinary generating function has reduced denominator $\prod_i (1 - \lambda_i u)^{d_i+1}$.

This is Lemma 6.1 of Part I, restated in Part II's notation (Part I writes E for the shift and t for the generating-function variable). The manuscript proves it again; its proof isolates each block by annihilating the others instead of using Chinese-remainder projectors, and is kept as a second route.

Second proof (manuscript 03). On a term $Q(N)\lambda^N$, the operator $E - \lambda$ gives $\lambda^{N+1}(Q(N+1) - Q(N))$. It lowers the degree by exactly one until the polynomial is zero. In contrast, $E - \mu$, for $\mu \neq \lambda$, preserves the degree and multiplies the leading coefficient by $\lambda - \mu$.

These observations also prove independence of distinct exponential blocks. To isolate one block in a purported zero sum, annihilate all other blocks by their respective powers of $E - \lambda_i$. The selected block remains nonzero unless it was zero initially. If a polynomial g has a zero of multiplicity e at λ_i , then its action on that block is nonzero for $e \leq d_i$ and zero for $e > d_i$. Hence each factor in (19.1) is necessary and their product suffices.

For the generating function, apply powers of $u d/du$ to $(1 - \lambda_i u)^{-1}$. A degree- d_i polynomial multiplier yields a pole of order exactly $d_i + 1$ at $u = \lambda_i^{-1}$. Distinct poles cannot cancel. This gives the stated reduced denominator. $\square$

Applying the lemma to Theorem 18.1 proves (14.11) under nonresonance. Without that hypothesis its labelled product still annihilates every term. Summing its exponents uses

$$\sum_{k=0}^t k(t-k) = \frac{t(t+1)(t-1)}{6}, \quad (19.2)$$

and gives (14.12). If two or more labels have the same rate, their sum has degree at most their largest degree, strictly less in order contribution than the sum of their separate $d + 1$ values. Thus any resonance strictly lowers the order below $\mathcal{R}_0$. This completes the proof of Theorem 14.2 and its corollaries.

In the nonresonant case the ordinary generating function is therefore

$$\sum_{N \geq 0} H_N(q)u^N = \frac{B_q(u)}{\prod_k (1 - \lambda_k u)^{d_k+1}}, \quad \deg B_q < \mathcal{R}_0, \quad B_q(0) = 1, \quad (19.3)$$

and the displayed fraction is reduced. A limiting specialization of this numerator and denominator need not remain reduced.

19.2 Comparison with the classical exponential bound

There is a useful elementary consistency check on the sharp formula. For $0 \leq k \leq t$,

$$1 + k(t-k) \leq \binom{t}{k}. \quad (19.4)$$

To see this, identify a k -subset of $\{1, \dots, t\}$ with a partition inside a $k \times (t - k)$ rectangle. Every size from 0 to $k(t - k)$ is attained: fill complete rows and then at most one partial row. There are therefore at least $k(t - k) + 1$ different partitions. Also

$$1 + \binom{m}{2} \leq 2^m, \quad 1 + \binom{\ell + 1}{2} = 1 + \ell + \binom{\ell}{2} \leq 2^\ell.$$

For nonnegative numbers a_j , $1 + \sum_j a_j \leq \prod_j (1 + a_j)$. Combining these inequalities, label by label and then summing, proves

$$\mathcal{R}_0 \leq 2^{m+\ell} \prod_i \sum_{k_i=0}^{t_i} \binom{t_i}{k_i} = 2^D. \quad (19.5)$$

Thus the order bound here is compatible with the classical one, while retaining the determinant cancellations that the larger bound ignores. For a single repeated root, summing $1 + k(t - k)$ gives $t + 1 + \binom{t+1}{3}$. Strictness for $t \geq 4$ already follows from $\binom{t}{2} > 1 + 2(t - 2)$ at $k = 2$.

19.3 Proof of the secondary Hankel factorization

We prove a general identity, then substitute the explicit leading coefficients of our determinant sequence.

Lemma 19.2 (A finite Hankel determinant of exponential polynomials). *Suppose $f(N) = \sum_{i=1}^b Q_i(N) \lambda_i^N$, with $\deg Q_i = d_i$, leading coefficient a_i , and $r_i = d_i + 1$. Put $R = \sum_i r_i$. The rates may coincide. Then*

$$\det_{0 \leq u, v < R} f(u + v) = (-1)^{\sum_i \binom{r_i}{2}} \prod_i (d_i! a_i)^{r_i} \lambda_i^{r_i(r_i-1)} \prod_{i < j} (\lambda_j - \lambda_i)^{2r_i r_j}. \quad (19.6)$$

Proof. Write $Q_i(N) = \sum_{h=0}^{d_i} a_{i,h} N^h$. Let $U_{u,(i,h)} = \lambda_i^u u^h$, $0 \leq h \leq d_i$, and let B be block diagonal, with i th block

$$(B_i)_{h,j} = \begin{cases} \binom{h+j}{h} a_{i,h+j}, & h+j \leq d_i, \\ 0, & h+j > d_i. \end{cases}$$

The binomial expansion of $Q_i(u + v)$ gives the exact factorization $(f(u + v)) = U B U^\top$. Each B_i is anti-triangular, so

$$\det B_i = (-1)^{\binom{r_i}{2}} a_i^{r_i} \prod_{h=0}^{d_i} \binom{d_i}{h}.$$

Lemma 17.1 gives $\det U$. The factorials simplify using

$$\left(\prod_{h=0}^{d_i} h! \right)^2 \prod_{h=0}^{d_i} \binom{d_i}{h} = (d_i!)^{r_i}.$$

Taking determinants proves (19.6) for distinct rates. The same identity holds for coincident rates by polynomial continuation in the λ_i , or directly because U then has repeated columns and both sides vanish. $\square$

Theorem 14.5 follows on taking the canonical labels $\mathbf{k}$ as the blocks of Lemma 19.2 and inserting (18.8). All its non-resonance factors are nonzero throughout the admissible root-multiplicity stratum. In particular, no hidden amplitude-zero locus can lower the maximal recurrence order.

Suppose, locally, exactly two rates meet, their difference vanishes to first order along a parameter curve, and no root itself reaches another root or an endpoint. Formula (14.16) then shows that the secondary determinant vanishes to exactly $2r_k r_j$ orders along that curve. This is a quantitative statement about rank loss, not just a test that some exceptional parameter exists.

20 Resonance and a root-free exact algorithm

20.1 The exact cancellation rule

Let Λ be the set of distinct numerical values among the rates. For $\lambda \in \Lambda$ define

$$Q_\lambda(N) = \sum_{k:\lambda_k=\lambda} P_k(N). \quad (20.1)$$

Some of these sums may be identically zero even though every summand has a nonzero leading coefficient.

Theorem 20.1 (All-parameter minimality rule). *At every admissible parameter choice, the minimal characteristic polynomial is*

$$\chi_q(X) = \prod_{\substack{\lambda \in \Lambda \\ Q_\lambda \neq 0}} (X - \lambda)^{1+\deg Q_\lambda}. \quad (20.2)$$

Before computing any cancellations, a valid smaller annihilator than the labelled product is

$$\Psi_q(X) = \prod_{\lambda \in \Lambda} (X - \lambda)^{1+\max\{d_k:\lambda_k=\lambda\}}. \quad (20.3)$$

Proof. Group (14.9) by numerical rate, discard zero polynomials, and apply Lemma 19.1. The degree of a sum is at most the maximum degree of its summands, proving (20.3). $\square$

Equation (20.2) is exact, but it is not a short classification of every possible resonant cancellation. Such a classification would require a more detailed description of the polynomials Q_λ . The next theorem instead gives a terminating algebraic procedure for each exact input, without constructing any spectral roots.

20.2 Why a finite prefix can be a proof

Theorem 20.2 (Certified finite-prefix recovery). *Let $h = (h_N)_{N \geq 0}$ be a nonzero sequence over a characteristic-zero field, known to satisfy a monic constant-coefficient recurrence of order at most R from $N = 0$ onward. Its minimal recurrence can be recovered from $h_0, \dots, h_{2R-1}$ as follows. For $r = 1, \dots, R$, solve*

$$h_{n+r} + \sum_{j=0}^{r-1} g_j h_{n+j} = 0, \quad 0 \leq n < R. \quad (20.4)$$

The first soluble system gives the unique minimal monic characteristic polynomial $X^r + \sum_{j < r} g_j X^j$.

Proof. Let $A(E)h = 0$ be a known-to-exist monic annihilator of order R , padding a lower-order one by a power of X if necessary. For any candidate g the residual sequence $u = g(E)h$ also satisfies $A(E)u = 0$, because polynomials in shift commute. If (20.4) holds, its first R values are zero. The monic recurrence then forces every subsequent value to be zero by induction. Thus every soluble system gives a genuine all-index annihilator, not merely a fitted prefix.

The actual minimal annihilator gives a solution for its order, so the first successful order is minimal. Two different monic annihilators of that degree would differ by a nonzero annihilator of smaller degree, which is impossible. The largest index used by the algorithm is $R - 1 + R = 2R - 1$. $\square$

The prior order bound is indispensable. Without it, an arbitrarily long matching prefix does not certify a recurrence for an arbitrary sequence. For (14.1), Theorem 14.2 supplies exactly the bound needed.

20.3 No factorization into individual roots is required

Suppose $q \in \mathbb{Q}[x] \setminus \{0\}$. Strip its powers of x and $x - 4$, and perform square-free decomposition on the remainder:

$$q(x) = \gamma x^m (x - 4)^\ell \prod_{t \geq 1} f_t(x)^t, \quad (20.5)$$

where the nonconstant f_t are monic, square-free and pairwise coprime. Let $b_t = \deg f_t$. Ordinary polynomial gcds with the derivative produce this decomposition; irreducible factorization and numerical root finding are unnecessary. The bound becomes

$$R = \prod_{t \geq 1} (t + 1)^{b_t} \left(\kappa + 1 + \frac{1}{6} \sum_{t \geq 1} b_t t(t - 1) \right). \quad (20.6)$$

Compute the first $2R$ determinants directly from their rational moment matrices and apply Theorem 20.2. This determines the true minimal polynomial even when the roots are complex, multiplicatively dependent, or not conveniently expressible by radicals.

All recovered coefficients lie in $\mathbb{Q}$. More generally, if the input coefficients lie in an exactly computable algebraic number field, the same linear systems work over that field. Solvability over its algebraic closure is equivalent to solvability over the field itself, by Gaussian elimination; uniqueness of the minimal monic polynomial gives descent. For arbitrary unspecified complex numbers, “algorithm” presupposes an exact representation with equality and field operations available. We do not assert an effective equality test for arbitrary complex oracles.

The transparent root-free implementation uses $O(R^4)$ field operations for generating the determinants and searching all orders with naive Gaussian elimination, apart from polynomial preprocessing and moment generation. This is an arithmetic-operation estimate, not a bit-complexity claim. The included fixed-size Christoffel path is much faster when the roots are already given exactly.

20.4 Algebraic rank-drop loci

The finite-prefix proof also turns exceptional orders into explicit polynomial conditions. Fix a degree $D \geq 0$ and consider nonzero polynomial multipliers of exactly that degree. The uniform all-index bound $S = 2^D$ follows from (19.5). For $1 \leq r < S$, form

$$\mathcal{H}_{S,r}(q) = (H_{n+j}(q))_{0 \leq n < S, 0 \leq j \leq r}. \quad (20.7)$$

Then

$$\text{ord}(H(q)) \leq r \iff \text{rank } \mathcal{H}_{S,r}(q) \leq r. \quad (20.8)$$

For the forward direction, the recurrence gives a column dependence. For the reverse direction, take a nonzero dependence, normalize its highest nonzero coefficient, and apply the residual argument of Theorem 20.2. Its degree is at most r .

Every entry $H_n(q)$ is a polynomial in the coefficients of q . Consequently the order-at-most- r locus is defined, within the degree- D coefficient space, by the vanishing of all $(r + 1)$ -minors of (20.7). It is relatively Zariski closed. On a fixed root-multiplicity stratum, S may be replaced by the sharper $\mathcal{R}_0$. Theorem 14.5 is an explicit factorization of the maximal-rank test on that stratum; lower-rank tests can involve further cancellation conditions after rates have already collided.

21 Worked examples and checks against the repository

21.1 Recovering the shifted linear multiplier

Take $q(x) = x^m(a + bx)$, with $b \neq 0$ and $a(a + 4b) \neq 0$. Then $\gamma = b$, $\ell = 0$, $s = 1$, $t_1 = 1$ and $c_1 = -a/b$. The two rates are $-bz$ and $-bz^{-1}$, so their sum and product are

$$-b(z + z^{-1}) = -b(c_1 - 2) = a + 2b, \quad (-bz)(-bz^{-1}) = b^2.$$

Their two degrees are both $\binom{m}{2}$. Theorem 14.2 gives exactly (13.1). At $a = 0$ the multiplier is bx^{m+1} , whose order is $\binom{m+1}{2} + 1$. At $a = -4b$ it is $bx^m(x - 4)$, whose order is $\binom{m}{2} + 2$. These endpoint results follow from the same theorem, without taking a singular limit of the two nonendpoint amplitudes.

Relation to Part I. In this subsection a and b are Part I's parameters, $q(x) = x^m(a + bx)$ is the multiplier of $D_N^{(m)}(a, b) = H_N(q)$, and $-z^{\pm 1}$ are Part I's roots $r_{\pm}$ of (6.1), so the two rates are Part I's $\lambda_{\pm}$ of (6.3). The paragraph is thus a second proof, by a different route, of the minimality in Theorem 2.2 and of the rows $a = 0$, $a = -4b$ and $a(a + 4b) \neq 0$ of Theorem 2.3; the row $b = 0$, $a \neq 0$ is the case $s = 0$ of Corollary 14.4 (an observation of this merge). It does not reprove Part I's coefficient formula, numerator degree, reciprocity or asymptotic constant. One further consistency check (an observation of this merge, not of the manuscript): for $q = x^m$, Theorem 18.1 gives the leading coefficient $B_0(m) (-1)^{\binom{m}{2}} \prod_{h < m} h! = 2^{\binom{m}{2}} \prod_{h < m} h! / \prod_{r < m} (2r)!$, which is Part I's h_m of (1.6), the leading coefficient of $H_N^{(m)}$.

21.2 A repeated fourth power with order fifteen

For

$$q(x) = (2x - 9)^4 = 16(x - \frac{9}{2})^4, \quad (21.1)$$

choose $z = 2$. The five rates, their exact degrees and their leading coefficients are as follows.

k	λ_k	d_k	$\text{lc}(P_k)$
0	1	0	1/729
1	4	3	8/81
2	16	4	-64/27
3	64	3	-2048/81
4	256	0	65536/729

The exact minimal characteristic polynomial is

$$\chi_q(X) = (X - 1)(X - 4)^4(X - 16)^5(X - 64)^4(X - 256). \quad (21.2)$$

The full expansion $H_N = \sum_{k=0}^4 P_k(N) \lambda_k^N$ is

$$\begin{aligned} P_0(N) &= \frac{1}{729}, \\ P_1(N) &= \frac{24N^3 + 204N^2 + 572N + 520}{243}, \\ P_2(N) &= -\frac{1728N^4 + 13824N^3 + 43344N^2 + 62784N + 35648}{729}, \\ P_3(N) &= -\frac{6144N^3 + 21504N^2 + 23552N + 10240}{243}, \\ P_4(N) &= \frac{65536}{729}. \end{aligned} \quad (21.3)$$

These polynomials may be obtained by (18.4), or by exact interpolation after the theorem has established their degrees and nonvanishing. The first determinant values are

$$1, \quad 3401, \quad 2480745, \quad 1026904425, \quad 327789036905, \quad 92503701028201.$$

No polynomial of degree 14 can annihilate this sequence: all five rates are distinct and every displayed top coefficient is nonzero. This nonexistence assertion is proved for the entire sequence, not inferred from the displayed six values.

21.3 Resonance can cancel an entire branch

For $q(x) = (x - 2)^2$, choose $z = i$. The three canonical sectors are

$$P_0(N) = \frac{i}{4}, \quad P_1(N) = N + 1, \quad P_2(N) = -\frac{i}{4}, \quad (21.4)$$

with rates $-1, 1, -1$. The two nonzero constant sectors cancel exactly, so

$$H_N((x - 2)^2) = N + 1, \quad \chi_q(X) = (X - 1)^2.$$

For completeness, the leading-coefficient formula gives the two constants and the coefficient 1 of N in P_1 . The condition $H_0 = 1$ then fixes its constant term. This already proves (21.4). Taking the maximum degree separately for each collided rate, without checking cancellation, would retain an unnecessary $X + 1$ factor.

A similar phenomenon need not involve repeated roots of q . Take

$$q(x) = (x - \frac{9}{2})(x + \frac{1}{2}), \quad z_1 = 2, \quad z_2 = -2. \quad (21.5)$$

The four canonical constants, in order $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, are $-1/15, -1/3, 1/3, 16/15$; their rates are $-1/4, -1, -1, -4$. Hence

$$H_N(q) = \frac{16}{15}(-4)^N - \frac{1}{15}(-\frac{1}{4})^N, \quad \chi_q(X) = X^2 + \frac{17}{4}X + 1. \quad (21.6)$$

The multiplicity profile permits generic order 4, the grouped annihilator has order 3, and exact cancellation lowers the true order to 2. Thus root multiplicities alone cannot determine the order on the resonance locus.

21.4 A nine-value proof of a resonant fourth-power formula

For $q = (x - 2)^4$ and $z = i$, the degree law and grouped annihilator give

$$(E - 1)^5(E + 1)^4H = 0. \quad (21.7)$$

The exact closed formula is

$$H_N((x - 2)^4) = \frac{(N + 1)(N + 2)^2(N + 3)}{12}. \quad (21.8)$$

Here is a finite proof using the already established all-index bound, rather than an extrapolation from data. The right side is a polynomial of degree 4, so it satisfies (21.7). The first nine values of both sides are

N	0	1	2	3	4	5	6	7	8
H_N	1	6	20	50	105	196	336	540	825

These determinant values follow from the 4×4 formula (15.8). They can be reproduced using only integers: write $u_{n,r} = p_n^{(r)}(2)/r!$, start with

$$(u_{0,r})_{r=0}^3 = (1, 0, 0, 0), \quad (u_{1,r})_{r=0}^3 = (1, 1, 0, 0),$$

and apply

$$u_{n+1,r} = u_{n,r-1} - u_{n-1,r}, \quad u_{n,-1} = 0.$$

Then $H_N = \det(u_{N+j,r})_{0 \leq r, j < 4}$. The difference between the two sides of (21.8) satisfies a monic recurrence of order 9 and has its first nine values zero. Induction proves that it is identically zero. Since the product polynomial has degree 4 and nonzero leading coefficient $1/12$, its minimal polynomial is $(X - 1)^5$. The generic order 15 has dropped to 5.

21.5 An irreducible rational input

No spectral representation is needed to run the exact algorithm. For $q = x^2 + 1$, the square-free profile is $b_1 = 2$, so $R = 4$. The root-free calculation gives

$$\chi_q(X) = X^4 - 5X^3 + 4X^2 - 5X + 1. \quad (21.9)$$

For $q = (x^2 + 1)^2$ the profile is $b_2 = 2$, and the order is 15. The archive records its full characteristic polynomial and exact input coefficients. These examples illustrate the intended use of the algorithm: its input is a rational coefficient list, not approximate complex roots or guessed multiplicative relations among them.

22 Exact asymptotics and stability of the leading sector

The expansion (14.9) is finite and exact. Consequently it contains more information than a leading asymptotic equivalence. Suppose a label $\mathbf{k}_*$ has strictly larger rate modulus than every other label. Put $\lambda_* = \lambda_{\mathbf{k}_*}$, $d_* = d_{\mathbf{k}_*}$, and let $M = \max_{\mathbf{k}} d_{\mathbf{k}}$. Choose $\rho < |\lambda_*|$ no smaller than the other rate moduli. Then

$$H_N(q) = \lambda_*^N P_{\mathbf{k}_*}(N) + O(\rho^N N^M). \quad (22.1)$$

When there are no other labels the error term is zero. In particular,

$$H_N(q) \sim L_{\mathbf{k}_*} N^{d_*} \lambda_*^N.$$

Every algebraic correction in inverse powers of N can be read from the finite polynomial $P_{\mathbf{k}_*}$; no infinite asymptotic series is needed for fixed parameters.

Corollary 22.1 (Repeated roots outside the interval). *Suppose all nonendpoint roots satisfy $c_i > 4$, and choose $z_i > 1$. Then the unique largest-modulus sector is $k_i = t_i$ for every i . Thus*

$$H_N(q) \sim L_{(t_1, \dots, t_s)} N^\kappa \left((-1)^{\ell+T} \gamma \prod_i z_i^{t_i} \right)^N. \quad (22.2)$$

No multiplicative-independence assumption is needed for this conclusion.

Proof. Reducing any k_i multiplies the modulus by a factor $z_i^{-2} < 1$. Hence the maximal sector is unique even if some of the smaller sectors resonate. At $k_i = t_i$ all $k_i(t_i - k_i)$ vanish, leaving degree κ . Its leading coefficient is nonzero by Theorem 18.1. $\square$

This separates two effects that are easy to conflate. Repeating a root outside the support changes the exponential growth rate and can greatly increase the recurrence order. It does *not* increase the power of N in the leading asymptotic term; that power comes entirely from the endpoint multiplicities. The repeated offendpoint roots instead create higher polynomial degrees in intermediate, subdominant sectors. For (21.1), for instance,

$$H_N((2x - 9)^4) \sim \frac{65536}{729} 256^N,$$

even though its 16^N sector has degree 4 and its recurrence has order 15.

The estimate is for fixed parameters. As roots approach 0, 4 or each other, the spectral gaps and the constants need not be uniform. A double-scaling limit cannot be justified by simply using (22.1) with moving parameters. That is one of the separate research directions listed below.

23 Verification, implementation, and formalization status

23.1 Independent exact calculations

The archive contains a reference implementation in Python 3.10 or later, using only the standard library. Every numerical calculation uses integers or exact rational numbers. Gaussian elimination uses pivoting; there is no assumption that a signed or resonant moment matrix is positive definite.

The implementation has two determinant paths. The first constructs the original matrix in (14.1) from Catalan numbers. The second computes jets from the polynomial recurrence (15.1) and uses the fixed-size Christoffel determinant. The paths share the low-level elimination routine, but neither uses the explicit sector leading constants. A third calculation interpolates sector polynomials at distinct rational spectral parameters and checks their degrees, leading constants and extra values. The secondary-Hankel factorization is tested by yet another determinant, built from the already computed sequence values.

The following are the actual results recorded by `code/verify.py` (shipped here as `code/02-root-collisions-verify.py`; its recorded run is `data/02-root-collisions-verification.json`, with console text `data/02-root-collisions-verification.txt`). Counts include simple and boundary instances; they measure executed identities, not independent universal theorems.

Executed check	Count
Endpoint-jet identities	304
Original moment determinant versus Christoffel determinant	1,554
Summed sector orders versus the multiplicity formula	12
Secondary-Hankel product factorizations, including resonances	15
Exact sector degrees and leading coefficients	48
Full exponential-polynomial evaluations	328
Nonresonant recurrence residuals	200
Nonresonant minimal-recurrence recoveries	9
Resonant recurrence residuals	204
Resonant minimal-recurrence recoveries	12
Inter-root resonance recoveries	3
Root-free multiplicity profiles	111
Root-free recurrence recoveries	3
Central-root fourth-power product evaluations	40
Total; all passed	2,843

The original-matrix comparisons cover 222 multiplier/scale choices at sizes $N = 0, \dots, 6$. They include roots below the support, in its interior, at both endpoints and above the support; multiplicities up to five

occur. The sector checks use twelve nonresonant patterns, including mixed endpoint and repeated-root blocks, with recurrence orders up to 26. The central fourth-power formula is checked through $N = 39$, although its finite proof in Section 21 needs only the first nine values.

The distinction between evidence and proof matters. A successful interpolation for twelve parameter choices does not prove Theorem 18.1. Its proof is the unique maximal-degree Cauchy–Binet selection and the three nonzero local determinants. Conversely, once a universal order bound is available, the finite certificate for a *particular* specialization is a rigorous proof by Theorem 20.2. The article uses both kinds of computation but does not conflate them.

23.2 The implementation interface

The module `code/catalan_collisions.py` provides the following principal entry points. (It is shipped here as `code/02-root-collisions-catalan_collisions.py`; §27 explains how to import it.) Polynomial arrays always use ascending powers.

Routine	Purpose
<code>direct_hankel</code>	Original moment determinant for rational coefficients.
<code>christoffel_values</code>	Fixed-size evaluation with rational roots and multiplicities.
<code>sectors</code>	Rates, exact degrees and leading constants for rational z_i .
<code>secondary_hankel_formula</code>	The product in (14.16).
<code>fit_sectors</code>	Exact polynomial interpolation when rates are distinct.
<code>multiplicity_profile</code>	Endpoint multiplicities and square-free root counts.
<code>minimal_recurrence</code>	Recovery from a prefix, given a proved order bound.
<code>recurrence_from_coefficients</code>	Root-free minimal recurrence for rational q .

The last routine is intentionally transparent, not optimized for large bounds. Its optional `max_bound` argument prevents an unexpected large computation. The mathematical theorem permits all complex roots; the reference implementation uses rational inputs, and does not purport to implement a general algebraic-number package.

23.3 A feasible formalization route

No Lean source has been compiled or checked for this article. The repository location does not confer formal status on the present proofs. A formalization should separate the argument into small algebraic layers.

First formalize normalized jets of (15.1) over a characteristic-zero field and their generating functions. Next prove the confluent Vandermonde identity, including the Euler-derivative row change, and formalize the polynomial-shift coefficient bound (17.1). The decisive reusable lemma is the maximal-degree Cauchy–Binet argument: it tracks one distinct list of derivative orders per spectral node and identifies the unique smallest such list.

The orthogonality and integral proof of Christoffel may be formalized analytically, but an algebraic alternative is attractive. One may define the moment functional by $L(x^n) = C_n$, prove its Gram identities for p_n directly from generating functions, and prove Christoffel by determinant multilinearity. This would avoid importing measure theory into the core recurrence theorem.

Finally, the shift-operator minimality lemma, the UBU^T secondary factorization and the finite-prefix recovery theorem are pure finite linear algebra over a field. They should be reusable outside Catalan determinants. An audit must distinguish imported classical facts from newly checked lemmas and must not replace a missing confluence proof with an admitted theorem hidden inside a computation.

24 Further research questions and proposed next steps

The questions below are precise continuations of this article. They are not all asserted to be unsolved in the published literature; a broader literature check is part of each proposed project.

24.1 Closed classification of cyclotomic resonances

For a single repeated root, classify the minimal recurrence as an explicit function of t and the order of z^2 when z is a root of unity other than ± 1 . The degree law reduces the task to finitely many grouped polynomials, but the cancellations can be substantial. The examples at $z = i$ show that counting distinct rates is not enough. A first target is a uniform closed form for $H_N((x-2)^t)$ for every t , together with a direct explanation of the parity effects, rather than a separate finite certificate for each t .

24.2 Pairwise resonance and cancellation order

Suppose exactly two labelled rates agree and all other rates are separate. Determine when their leading coefficients cancel, and how many further coefficients can cancel. Theorem 18.1 makes the first test explicit, while (16.6) determines all later tests. The aim is a structural criterion in the root parameters, not just the output of Gaussian elimination. The distinct-root example (21.5) is a small initial case.

24.3 Sector polynomials and symmetric functions

The degree $k(t-k)$ is the area of a $k \times (t-k)$ rectangle, and the proof uses a mixed confluent Vandermonde. Seek a Schur-function or factorial-Schur description of the entire P_k , not only its leading coefficient. Useful outputs would be exact factorizations, coefficient sign rules for real roots outside $[0, 4]$, and formulas that remain stable as two parameters approach one another. Such a description could also identify an earlier equivalent formulation of the multiplicity law.

24.4 Uniform endpoint and root-collision limits

A concrete endpoint scaling is

$$c_N = 2 + 2 \cosh(\tau/N), \quad q_N(x) = (x - c_N)^t.$$

Determine the limit, locally uniformly in τ , of

$$\frac{H_N(q_N)}{(-1)^{tN} N^{t(t+1)/2}},$$

including its value at $\tau = 0$. The analogous left-endpoint scaling $c_N = 2 - 2 \cosh(\tau/N)$ should use the exponent $t(t-1)/2$. For $t = 1$ the branch formula already reduces these limits to elementary hyperbolic functions. The general problem asks for a uniform confluent determinant limit and error bounds. A second version lets two distinct nonendpoint roots approach at a rate of order N^{-1} .

24.5 Other eventually constant Jacobi recurrences

Krattenthaler's recurrence theorem covers many moment families beyond Catalan numbers [3]. Extend the exact degree analysis to a three-term recurrence whose coefficients are eventually constant. Away from branch points, the main new issue is a zero of one branch amplitude caused by the finite initial perturbation. Classify how its vanishing order changes the jet degrees and the noncancellation matrix. This would distinguish universal collision behavior from behavior tied to the Catalan boundary condition $p_1 = x - 1$.

24.6 Rational multipliers and finite poles

Replace $q(x)$ by a rational function whose poles avoid the support. The determinant identities underlying Uvarov transformations provide a starting point [8]. The target is an exact analogue of the sector-degree formula when evaluation and Cauchy-transform blocks occur together. One must first specify which normalizing factors are removed and which recurrences hold from the initial index rather than only eventually.

24.7 Coefficient-space equations and arithmetic reductions

Eliminate the spectral variables from (14.16) to obtain small defining equations for maximal-order failure directly in the coefficients of q . The finite minors in (20.7) are correct but can be very large. Seek sharper degrees, irreducible components and efficient membership tests. For monic integral multipliers, a related problem is to classify primes at which the minimal recurrence drops after reduction. The factorials in the leading constants show why characteristic-zero noncancellation cannot simply be reused unchanged in small characteristic.

24.8 Fast certified algorithms and formal verification

Combine the proved order bound with fast minimal-recurrence algorithms, but reduce the cost of obtaining the initial determinants as well. A useful target is an output-sensitive exact algorithm with bit-complexity bounds in D , the coefficient heights and the final recurrence order. Separately, formalize the maximal-degree determinant lemma and the finite-prefix certification theorem first: these are the shortest reusable path from the present article to a machine-checked result.

25 Conclusion

The selected repository extension has a uniform answer at the level of exponential sectors. A root of multiplicity t_i away from 0, 4 contributes the exact degree $k_i(t_i - k_i)$ to its k_i th branch sector. The two endpoints contribute different triangular numbers. The leading coefficient is a nonzero product, so distinct sectors cannot disappear at nonresonant parameters. This proves the sharp order formula (14.12), with cubic rather than exponential complexity for a single repeated root.

The secondary-Hankel factorization identifies the maximal-order exceptional locus exactly. Resonances may cause further cancellations, but they do not obstruct an exact answer for a specified rational or algebraic multiplier: a proved order bound turns finitely many determinant values into a complete minimal-recurrence certificate. The remaining research questions concern structural descriptions of those cancellations, uniform collision limits, broader moment families and formalization, not gaps in the stated multiplicity proof.

26 Normalization and sign audit

There are four easy ways to obtain an incorrect statement in this problem. This section (Appendix A of the manuscript) records the conventions that prevent them.

Size versus last index. Throughout, N is the determinant size. In a convention that indexes rows by $0 \leq i, j \leq n$, one must substitute $N = n + 1$. The initial term $H_0 = 1$ is part of the recurrence sequence used here.

The Christoffel sign. The product in the multiplier is $\prod (x - a_u)$, whereas the mixed Vandermonde supplies $\prod (a_u - x)$. This contributes $(-1)^{ND}$. The m zero-endpoint rows add $(-1)^{mN}$ when their factors are removed. The remaining sign is therefore $(-1)^{(\ell+T)N}$, exactly the sign in (14.6).

Normalized versus unnormalized jets. The rows of M_N are $p_{N+j}^{(r)}(a)/r!$. Thus Δ_q contains only cross-root differences. Extra products of $r!$ in its denominator would double-count confluence. The factorials appearing later in B_0, B_4, B_z and $\mathcal{V}$ have separate origins and must not be removed by changing the jet convention halfway through the proof.

Two different kinds of collision. A collision of roots of q changes the factorization profile and must be treated by normalized jets. A collision of rates λ_k can occur even when all roots of q are distinct, and must be treated by adding their polynomial coefficients. These are logically different operations. The first creates the degree law; the second can reduce the actual recurrence order.

As a final check, $H_N(uq) = u^N H_N(q)$. The determinant on the left of (14.16) therefore scales by $u^{\mathcal{R}_0(\mathcal{R}_0-1)}$. On its right, the rate powers and rate differences contribute

$$\sum_k r_k(r_k - 1) + 2 \sum_{k < j} r_k r_j = \mathcal{R}_0(\mathcal{R}_0 - 1),$$

while the sector leading coefficients do not depend on the overall scale γ . The scaling agrees exactly.

27 Reproducing the package

This section is Appendix B of the manuscript, adapted to the shipped file names. As delivered, the package was reproduced from its top directory by

```
python3 code/verify.py --out data
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the recorded numerical outputs; the second builds the PDF. There are no third-party Python dependencies. *In this report these commands must not be run in place.* Here `code/verify.py` and `data/` belong to Part I, so the first command would run Part I's suite and rewrite Part I's data; the shipped verifier `code/02-root-collisions-verify.py` still imports its sibling module under the delivered name `catalan_collisions`, which is not importable under the shipped name, and its default output directory `data` would overwrite Part I's `data/verification.json` and `data/verification.txt`. The delivered Makefile, shipped as `code/02-root-collisions-Makefile`, has the same defect: its `check` target runs `code/verify.py --outdata`, which here is Part I's verifier. Restore the delivered names in a scratch copy instead (POSIX shell, run from the report directory):

```
W=/path/to/scratch/root-collisions
mkdir -p "$W/code" "$W/recorded"
for f in code/02-root-collisions-*.py; do
  cp "$f" "$W/code/${f#code/02-root-collisions-}"; done
for f in data/02-root-collisions-*; do
  cp "$f" "$W/recorded/${f#data/02-root-collisions-}"; done
cd "$W" && python3 code/verify.py --out data
```

The run writes seven files to `$W/data`, which can be compared with the recorded copies in `$W/recorded`; on 28 September 2026 this recipe passed all 2,843 checks, and the seven files agreed with the recorded ones apart from Windows line endings. The following root-free example, run in the same scratch directory, needs no spectral factorization:

```
import sys
```

```

sys.path.insert(0, "code")
from catalan_collisions import recurrence_from_coefficients

chi, bound = recurrence_from_coefficients([1, 0, 1], max_bound=16)
print(bound) # 4
print(chi)    # ascending coefficients: 1, -5, 4, -5, 1

```

The main data files are shipped in `data/` with the prefix `02-root-collisions-` (delivered without it):

- `verification.json`, `verification.txt`: the recorded run of Section 23;
- `sectors.json`: the twelve nonresonant patterns, with sector polynomials, leading constants and characteristic polynomials;
- `resonances.json`: exact minimal recurrences at resonant parameters;
- `rational_input_examples.json`: root-free rational-coefficient examples;
- `central_root_certificate.json`: the nine-value certificate of (21.8);
- `fourth_power_sequence.csv`: exact values of $H_N((2x - 9)^4)$.

The code is shipped in `code/` with the same prefix (`verify.py`, `catalan_collisions.py` and the delivered root `Makefile`), and the provenance notes, delivered as `notes/provenance.md`, as `02-root-collisions-provenance.md` at the report root. Rational values are stored as exact strings, for example `"8/81"`. Arrays representing polynomials use increasing powers. Sector polynomials are polynomials in the determinant size N , not generating-function numerators.

A The precise correction to the printed conjecture

This appendix distinguishes two kinds of index. The source uses a size- $(n + 1)$ determinant. We write $N = n + 1$. To avoid silently identifying incompatible shift conventions, write s for the source's symbol in its displayed coefficient formula, and reserve m for our actual Catalan shift.

For $s \geq 2$ and $0 \leq k \leq n + 1$, the coefficient expression printed in Conjecture 2 of [1] can be written as

$$T_{n,k,s} = \frac{C_s}{2s-2} \binom{s+k-2}{s-2} \binom{n+k+2s-2}{2k+2s-3} \times \prod_{j=0}^{\lfloor (s-1)/2 \rfloor - 1} \frac{\binom{2n+2s-2j-1}{2s-4j-5}}{\binom{2s-2j-1}{2s-4j-5}} \prod_{j=0}^{s-3} \frac{2s-j-2}{n+k+2s-j-2}. \quad (\text{A.1})$$

Only the natural, nonambiguous range $s \geq 2$ is needed here. Empty products have value 1.

Proposition A.1 (Correct shift). *For $N \geq 1$, $s \geq 2$, and $0 \leq k \leq N$,*

$$T_{N-1,k,s} = c_{N,s-1,k}. \quad (\text{A.2})$$

Consequently, the polynomial using $T_{N-1,k,s}$ is $D_N^{(s-1)}(a, b)$, not $D_N^{(s)}(a, b)$. Equivalently, to use that printed expression for an actual shift $m \geq 1$, replace its shift symbol by $s = m + 1$.

Proof. Taking a ratio of successive terms in (A.1) and cancelling factorials gives

$$\frac{T_{n,k+1,s}}{T_{n,k,s}} = \frac{(n+1-k)(n+k+s+1)}{2(k+1)(2k+2s-1)}. \quad (\text{A.3})$$

This agrees with (2.3) after $N = n + 1$ and $m = s - 1$. It remains to match the initial coefficient. Putting $n = N - 1$, $k = 0$, and simplifying the factorials gives

$$T_{N-1,0,s} = C_s \frac{(N+s-1)!}{N!s!} \prod_{j=0}^{p-1} \frac{\binom{2N+2s-2j-3}{2s-4j-5}}{\binom{2s-2j-1}{2s-4j-5}}, \quad p = \left\lfloor \frac{s-1}{2} \right\rfloor. \quad (\text{A.4})$$

At $N = 1$ this is $C_s = H_1^{(s)}$. Its ratio at successive N is

$$\begin{aligned} \frac{T_{N,0,s}}{T_{N-1,0,s}} &= \frac{N+s}{N+1} \prod_{j=0}^{p-1} \frac{(2N+2s-2j-1)(2N+2s-2j-2)}{(2N+2j+4)(2N+2j+3)} \\ &= \frac{(2N+2s)!(2N+1)!}{(2N+s)!(2N+s+1)!}. \end{aligned} \quad (\text{A.5})$$

In the product, the numerator factors form a consecutive interval of integers, as do the denominator factors; the last equality holds for either parity of s . The shifted-product formula gives the identical ratio

$$\frac{H_{N+1}^{(s)}}{H_N^{(s)}} = \prod_{i=1}^{s-1} \frac{(2N+2i+1)(2N+2i+2)}{(2N+i+1)(2N+i+2)} = \frac{(2N+2s)!(2N+1)!}{(2N+s)!(2N+s+1)!}.$$

Induction in N therefore proves $T_{N-1,0,s} = H_N^{(s)}$. The ratio (A.3) then proves (A.2) for every k . $\square$

The discrepancy is visible in the smallest nontrivial instance. With $N = 1$ and $s = 2$, the printed expression is

$$T_{0,0,2}b + T_{0,1,2}a = 2b + a,$$

whereas the determinant with the stated shift $s = 2$ is

$$D_1^{(2)}(a, b) = aC_2 + bC_3 = 2a + 5b.$$

At $a = b = 1$, these are 3 and 7. The expression $a + 2b$ is exactly $D_1^{(1)}(a, b)$, so the correction is systematic rather than an isolated numerical exception. The original PDF page was inspected as well as its HTML rendering; this is not an artifact of an HTML conversion. The denominator-exponent statement is separate and uses the actual shift m correctly.

B Compact reference and archive conventions

Symbol	Meaning
N	Determinant size, with $D_0 = 1$
m	Actual Catalan shift in $aC_{i+j+m} + bC_{i+j+m+1}$
$H_N^{(m)}$	Pure shifted Catalan determinant
d, e, K	$\binom{m}{2}$, $\binom{m}{2} + 1$, and $(m - 1)^2$
P_m	Uniform numerator, not automatically reduced at exceptions
q	$1 - (a + 2b)t + b^2t^2$
$M_{m,\ell}$	Minor with column ℓ deleted from the binomial block
$Q_m(N; r)$	Signed generating polynomial of those minors
$r_{\pm}$	Roots of $r^2 - (a/b + 2)r + 1$
$\lambda_{\pm}$	$br_{\pm}$, the characteristic roots in determinant size
s in Appendix A	Source shift symbol, equal to actual shift plus one

[Added 28 September 2026, batch 39: this table covers Part I. Part II reuses several of these letters with other meanings, among them q , s , ℓ , d , K , P , M and r ; its symbols are listed in Table 1, and its data files in Section 27.]

The JSON files use ascending polynomial powers. In `coefficient_rows.json`, position k is the coefficient of $a^k b^{N-k}$. In `minor_polynomials.json`, position j is the coefficient of N^j . In `two_branch_polynomials.json`, the arrays `P_plus` and `P_minus` describe the polynomial multipliers of 4^N and 1^N for $(a, b) = (1, 2)$; they are not the ordinary-generating-function numerator P_m .

The source audit records exact URLs, source versions, and search limitations. The tests establish reproducibility of the computation, not priority or formal verification. The archive intentionally contains no third-party article PDFs, no font files, no compiler auxiliaries, and no Python bytecode.

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Online sources checked on 19 September 2026 (Part I) and 28 September 2026 (Part II). All theorem numbering in the body refers to the present manuscript, unless a source is explicitly named.